Lucas corrections and elimination of arithmetic witnesses

Explicit Diophantine forms for Con(ZFC)

Prepared for Vladimir Reshetnikov and the ProveIt repository
Research, derivations, code and independent scoped reviews with AI assistance

25 October 2026
Comparison snapshot: `3ecd6a6cf53d47519228328567f45fb7ee9781d9`

Abstract

We give two new arithmetic transformations and a family of explicit polynomial projections for the Diophantine consistency problem. A second-order Lucas correction makes the previously obstructed scale $Y = qs$ usable in the compact native/Pell construction. It yields one equation in ten strictly positive integer variables, of exact degree 304, evaluated in 87 charged binary arithmetic operations. A separate sign-recovery argument combines two front nonnegativity certificates in the direct C222 construction. Its principal quartic has 248 unrestricted integer variables, 5,321 collected monomials, and coefficient height 128,446, improving its matched 252-variable parent without increasing degree or height. Acyclic monic elimination then gives, among other points, 153 unrestricted integers at degree 8, 125 at degree 16, and 102 at degree 256. The complete octic has 188,579 monomials and height 750,273,658,864.

For each delivered equation, the new ordinary arithmetic proof, composed with the identified inherited refutation calibration, proves in ZFC that absence of solutions in its stated domain is equivalent to syntactic Con(ZFC). The archive supplies literal equations, all fixed coefficient bytes and their reconstruction, restoration certificates, independent exact checkers, and complete printed specifications of the principal compact polynomial and the projected integer family. Global minimality and a new end-to-end proof-assistant formalization are not claimed.

Contents

1 The result, its domain, and its comparison boundary **3**
1.1 Three useful outcomes rather than an unspecified optimum . . . 3
1.2 What improves on the inspected sources . . . 3
1.3 The exact inheritance boundary . . . 4

2 A second-order Lucas correction: ten positive variables, degree 304 **5**
2.1 The complete polynomial . . . 5
2.2 Roots and early ranks without assuming source signs . . . 6
2.3 Absolute transport at the smaller scale . . . 7
2.4 The corrected numerator and all possible rank signs . . . 7
2.5 The second-order index lemma . . . 8
2.6 Projection and restoration of the native source . . . 9
2.7 Construction of positive witnesses . . . 9

2.8 Exact degree and validation 10

3 A direct quartic from one certificate for two front signs 11
3.1 Literal source change and inherited interface 11
3.2 The signed-base extension 12
3.3 Both front branches have the needed size bound 13
3.4 Recovering the radix before resolving the branch 13
3.5 The exact factorial digit chooses the positive branch 14
3.6 Completeness and the parity restriction 15
3.7 Exact endpoint and the logical calibration chain 16

4 Trading degree for fewer direct proof coordinates 16
4.1 The resulting explicit family 17
4.2 Exact degree and coefficient bounds 18
4.3 Verification and the consistency theorem 19

5 Verification, reproduction, and use of the archive 19
5.1 Files that define the result 20

6 What the new ideas change, and what remains open 20

A Complete fixed coefficient recipe for the compact equation 21

B All residuals of the principal direct quartic 23

C Complete monic-elimination pivot tables 35
C.1 225 integer variables, degree 6 35
C.2 153 integer variables, degree 8 35
C.3 144 integer variables, degree 12 35
C.4 125 integer variables, degree 16 36
C.5 117 integer variables, degree 32 36
C.6 111 integer variables, degree 64 37
C.7 102 integer variables, degree 256 37
C.8 94 integer variables, degree 65,540 38

1 The result, its domain, and its comparison boundary

The objective is a fixed polynomial P and a specified domain D such that

$$\mathrm{ZFC} \vdash \left(\neg\exists \mathbf{x} \in D\ P(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con(ZFC)}. \tag{1}$$

Here Con(ZFC) denotes the arithmetical assertion that there is no finite syntactic refutation in the repository's fixed recursive presentation of ZFC. Equivalence of reasonable effective proof presentations can be proved by the corresponding syntactic translations. The present paper uses the inherited presentation consistently and does not select coefficients on the basis of whether it is consistent.

The proof strategy establishes the positive existential statement first: an allowed zero exists if and only if a finite refutation exists. Negating that equivalence gives (1). No assumption of consistency is used to prove the equivalence, and this paper does not establish either its unsolvability side or its consistency side.

1.1 Three useful outcomes rather than an unspecified optimum

Different measures of simplicity conflict. An expression with a small number of unknowns can hide large fixed coefficients; expanding an expression can change its apparent length without changing the polynomial; eliminating an arithmetic witness can increase degree. We therefore give a primary compact result and a separate family with small explicit circuit numerals.

Construction	Unknowns	Degree	Main feature
Compact Lucas correction	10 positive	304	87 binary operations; the complete coherent fixed native tuple is supplied.
Principal direct quartic	248 integer	4	5,321 terms; height 128,446; direct C222 proof calculus.
Direct octic projection	153 integer	8	Entire expansion supplied: 188,579 terms, height 750,273,658,864.
Moderate-degree projection	125 integer	16	Literal circuit and integral restoration; coefficient height below 2^{80}.
Smaller-dimension projection	102 integer	256	Literal circuit and integral restoration; coefficient height below 2^{691}.

"Positive" means strictly positive, and "integer" means unrestricted integer. These domains are part of the theorem. The straight-line abbreviations in every displayed equation are definitions, not extra existential variables. The integer projections have only addition, subtraction, multiplication and fixed integer constants. Acyclic restoration is a finite polynomial construction, not an invocation of a universal Diophantine polynomial with unspecified parameters.

For the compact equation, each live binary addition, subtraction and multiplication is charged once; a square and multiplication by a fixed constant each count as one multiplication. This is arithmetic circuit cost, not bit complexity. The largest of its four stored fixed coefficient ports has 592,884 bits. In the direct projection circuits the largest literal constant is 256; this does *not* bound the coefficients after expansion. The separate height columns refer to those collected coefficients.

1.2 What improves on the inspected sources

All comparisons are to the pinned repository snapshot above, including the eight incoming archives recorded in `SOURCE_LEDGER.json`. The latest incoming material matters: comparing only against an

older maintained page would overstate the improvement. The inspected compact sources already contain 75/626, 76/464 and 77/460 operation/degree points, as well as the lower-degree points below [4, 3, 8, 2].

Ten-positive construction	Operations	Degree	Status
Centered residues	75	626	Prior comparison
Centered residues	76	464	Prior comparison
Centered residues	77	460	Prior comparison
Unshifted doubled product square	84	328	Prior comparison
Graph-digit center	87	323	Prior comparison
Sharp graph-digit center	88	318	Prior comparison
Second-order Lucas correction	**87**	**304**	This paper
Uncentered transport	89	288	Prior comparison
Lower-degree native form	121	246	Prior comparison

Thus the new 87/304 point improves both displayed graph-digit points in the cost–degree comparison, with the same ten-positive domain. It does not dominate every earlier compact point. The meaningful mathematical change is that a scale explicitly rejected in the inspected denominator-engineering notes becomes possible by correcting the index congruence [5]. No literature-wide priority claim is inferred from that repository comparison.

The direct quartic's matched parent has 252 unrestricted integer variables, 182 residuals, 5,340 terms and height 128,446 [7]. The new 248-variable principal improves variable and term counts while retaining the other metrics. Two further quartics use 246 and 245 variables with different height/sparsity tradeoffs. These counts should not be presented as global records for unrestricted integer Diophantine consistency: the repository already contains much smaller native-tuple forms [2]. The distinction here is the direct-calculus route, its explicit modest coefficients, and the complete algebraic maps.

1.3 The exact inheritance boundary

The compact contribution changes the arithmetic front of the existing native/Pell route. It retains the complete directly indexed BFS943 tuple and its original native/refutation calibration. Its source data ultimately descend from the existing machine/tile encoding; this paper does not claim that compact route is independent of K215. In contrast, the direct quartic and its higher-degree projections encode the C222 numerical proof calculus. Their arithmetic does not simulate K215 and does not use Jones's universal polynomial or Sun's reduction.

For the direct route the inherited final bridges are

`KEndpoint.c222_iff_zf_refutation`,

`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

The first theorem is about the machine-independent calculus even though its source file also contains machine endpoint theorems. The second is the constructibility-based equiconsistency bridge from ZF to ZFC. The delivered source closure retains these interface files and identifies their upstream dependencies. It is not a new standalone build of all transitive Lean imports.

The new contributions are ordinary unbounded mathematical proofs plus finite exact algebraic certificates. Finite checking authenticates the literal polynomials, restorations, degree witnesses, coefficient bytes and operation counts. It does not replace the Pell, CRT, native-compiler or syntactic calibration theorems. A claim of a new end-to-end Lean theorem, or a new internal-ZFC proof term, would therefore be stronger than this deliverable. All "independent reviews" supplied here are separate checks within an AI-assisted workflow; they are not outside human peer review.

2 A second-order Lucas correction: ten positive variables, degree 304

The principal result of this section is a fixed integer polynomial in ten strictly positive integer variables, with exact degree 304 and an evaluator using 87 binary arithmetic operations. The arithmetic change repairs a specific failed direction in the latest inspected submissions: with the smaller scale $Y = qs$, the elementary first-Pell congruence visibly supplies only q^3, whereas the unknown rank is bounded by q^4. Retaining the next term of the Lucas polynomial supplies enough precision to recover the index. The full denominator and the omitted native divisibility are then recovered in that order.

This is a new step within the inherited native/Pell encoding. The proof below composes it with the explicitly retained ordinary calibration of that encoding to finite syntactic ZFC refutability. The complete formula, fixed coefficient bytes and their finite reconstruction are supplied. No new end-to-end proof-assistant object or proof of consistency is claimed.

2.1 The complete polynomial

The only variables are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The four fixed integers β, K_0, M_C, M_F are the entire directly indexed BFS943 tuple of the retained parent. They satisfy

$$\beta \geq 31 \text{ odd}, \qquad K_0 > 0, \qquad 0 < M_C < \beta < M_F < 2\beta.$$

The largest stored port has 592,884 bits. These are fixed coefficients, not additional existential variables. Their complete finite recipe is retained separately from the mathematical abbreviations below.

Define the following polynomial expressions:

$$h = \beta J, \quad q = h + 1, \quad C = q - F - \alpha, \quad X = wq, \quad Y = qs,$$
$$k = \eta + \zeta, \quad c = kY + \eta, \quad A = Y(X + 1) + 1, \quad \Delta = A^2 - 1, \quad H = 4A - 5,$$
$$v = XY^2, \qquad E = 1 + v(v + 1)k^2,$$
$$R = [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \qquad B = (K_0 + w)C - F,$$
$$U = ic^2, \quad a = \Delta U^2 E, \quad F_0 = 1 + a, \quad G = EF_0,$$
$$F_* = 1 + 4aF_0, \quad Q = \Delta(F_* - 1), \quad b = cT - R, \quad V = F_* b - c, \quad K = X(3c - 2X) + 2.$$

The new corrected numerator is

$$e_6 = 6(k - R - 1) - v(R + 1)(R - 1)(R + 3). \tag{2}$$

For the principal equation put $L = bV$, and define

$$D_6 = 6q^6, \qquad M = hD_6H,$$
$$Z = hD_6[H(n - LB) + K] - H[D_6B + h(e_6 - 6q^4 s)],$$
$$R_G = M^2 G.$$

The requested single equation is

$$\boxed{P = Q(Z^2 - R_G V^2) + (R_G - Z^2) = 0.} \tag{3}$$

Every named expression is a polynomial abbreviation. No division, factorial, binomial coefficient, variable exponent or implicit additional witness occurs in the equation. Its literal evaluator is `equation_bV.json`.

Replacing L by V^2 gives a control equation with degree 410 in 86 operations. It is useful for separating the index correction from the sharper transport estimate, but is dominated by earlier cost–degree points and is not claimed as a frontier improvement.

Theorem 2.1 (Second-order compact consistency equation)**.** *For the complete fixed coefficient tuple just specified, the ordinary arithmetic proof below, composed with the retained native/refutation calibration, establishes*

$$\mathrm{ZFC} \vdash \left(\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{10}\ P(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con(ZFC)}.$$

The principal polynomial has exact degree 304*. Its supplied evaluator has* 53 *multiplications,* 18 *additions and* 16 *subtractions, for* 87 *binary operations in total, including squares and multiplication by fixed numerals.*

For the proof only, write

$$t = \frac{B}{h}, \quad W = 1 + hL, \quad \kappa = \frac{K}{H}, \quad \sigma = \frac{s}{q^2}, \quad \bar{e} = \frac{e_6}{6}, \quad \rho = \frac{\bar{e}}{q^6}, \quad z = \frac{Z}{M}.$$

These rational quantities need not initially be integral. Direct polynomial identities give

$$P = (Q-1)Z^2 - M^2G(QV^2-1), \qquad z = n + \kappa + \sigma - Wt - \rho. \tag{4}$$

In particular the appearance of $1/6$ in the proof does not introduce a rational-coefficient gate: the circuit computes the integer polynomial e_6.

2.2 Roots and early ranks without assuming source signs

Positivity gives $q \geq 32$, $X, Y \geq q$, $k \geq 2$, $c > kY$, $A \geq 3$, $U \geq c^2$, $F_* > 2c$, $Q > 1$ and $M > 0$. For every integer b, $V = F_*b - c$ is nonzero and $|V| > 1$. The exact identities

$$F_* = (1+2a)^2, \qquad Q = 4\Delta^2U^2G$$

show that G is coprime to both $Q-1$ and QV^2-1. At a zero of P, the first identity in (4) yields $(Q-1)z^2 = G(QV^2-1)$. Every prime valuation of G is even, so $G = g^2$ with $g > 0$ integral. Since $\gcd(E, F_0) = 1$, its positive factors are separately squares:

$$E = \tau^2, \qquad F_0 = f_0^2, \qquad g = \tau f_0.$$

The positive rational $y = |z|/g$ satisfies

$$QV^2 - (Q-1)y^2 = 1. \tag{5}$$

The adjacent-Pell descent forces y integral. To recall its short proof, write $y = r/m$ in lowest terms. Then $m^2 \mid Q-1$. If $u = |V| > 1$, then $u < y < Qu/(Q-1)$, and the norm-preserving transformation

$$u' = (2Q-1)u - 2(Q-1)y, \qquad y' = (2Q-1)y - 2Qu$$

has a nonzero integer first coordinate with $|u'| < u$ and preserves the reduced denominator m of its second coordinate. Reflection and descent reach $u = y = 1$, forcing $m = 1$. Thus $z = \varepsilon g y$ is integral, where $\varepsilon \in \{-1, 1\}$.

Set $f = 1 + 2a$ and $I = 2Ug$. Pell doubling gives

$$f^2 - \Delta I^2 = 1, \qquad Q = (\Delta I)^2, \qquad c^2 \mid I, \qquad V = f^2(cT - R) - c. \tag{6}$$

The retained signed all-index folding theorem applies to (5)–(6), $c \leq I$ and $V \equiv -c \pmod f$. It requires no sign for V or R and gives $c = \psi_A(p)$ for a positive odd p. Here $\psi_A(0) = 0$, $\psi_A(1) = 1$ and $\psi_A(j+2) = 2A\psi_A(j+1) - \psi_A(j)$. The first norm gives

$$k = 2\psi_{2v+1}(n_0), \qquad n_0 \geq 1.$$

If $n_0 = 1$, then $k = 2$, $\eta = \zeta = 1$, and $c = 2Y + 1$ lies strictly between the first two main ordinates 1 and $2A$. This is impossible. Consequently

$$n_0 \geq 2, \quad k \geq 8v + 4, \quad c > kY > 8XY^3 \geq 8q^4, \quad c > X, \quad K > 0. \tag{7}$$

No sign of B, C, R, b, V or $\bar{e}$ has been assumed.

2.3 Absolute transport at the smaller scale

If $b = 0$, equation (5) would give $y^2 - c^2 = (c^2 - 1)/(Q - 1)$ strictly between zero and one. Hence $b \neq 0$, $\operatorname{sgn} V = \operatorname{sgn} b$, and $|V| > |b|$. With $D_0 = \Delta U^2$, one has

$$\frac{a^2}{g^2} = \frac{D_0 a}{1 + a} > \frac{D_0}{2} > 9.$$

Thus $a > 3g$, $|V| > 3g$ and $y < 3|V|/2$. In particular $gy < V^2/2$ for the control.

For $L = bV$, the third main index must also be excluded. Suppose $c = 4A^2 - 1$. If $n_0 \geq 3$, then $k > 32v^2$, while $A < 2XY$ gives

$$c < 16X^2Y^2 < 32X^2Y^5 < kY < c,$$

a contradiction. Thus $n_0 = 2$ and $k = 8XY^2 + 4$. Put $u = X + 2 - 2Y$. Direct expansion gives

$$\eta = 4Y^2(Xu + 1) + 8Y(X + 1) + 3 - 4Y.$$

For $u \leq -1$ this is negative, using $Y(X - 1) > 2(X + 1)$. For $u \geq 2$, subtracting k leaves at least $4Y^2 + 8Y(X + 1) - 4Y - 1 > 0$. Both contradict $0 < \eta < k$. The remaining $u = 0, 1$ would make $2Y - X$ equal to 2 or 1, contradicting $q \mid X, Y$ and $q \geq 32$. Therefore $p \geq 5$ and $c > c_3 := 4\Delta + 3$.

Put $S = \Delta I$, so $S^2 = Q$. Pell classification of (5) gives $|V| = \chi_S(j)/S$ for a positive odd j. Index one gives $|V| = 1$ and is excluded. At index three, $|V| = 4Q - 3$ would give one of

$$F_*(b - 4\Delta) = c - c_3, \qquad F_*(b + 4\Delta) = c + c_3.$$

Both right sides lie strictly between zero and $2c < F_*$, impossible. Hence $j \geq 5$, $|V| \geq 16Q^2 - 20Q + 5 > 8Q^2$, and

$$|b| > \frac{8Q^2 - c}{F_*} > 4F_* > 4g.$$

It follows that $L = bV = |b||V| > 0$ and $gy < 3L/8 < L/2$. For either transport choice we have established

$$L > 0, \qquad gy < L/2, \qquad |b|/q^6 < L/2. \tag{8}$$

2.4 The corrected numerator and all possible rank signs

The new correction needs a fresh sign argument: its cubic changes sign at negative ranks. Set $e_0 = k - R - 1$ and $B_R = (R + 1)(R - 1)(R + 3)$. Since $b - e_0 = cT - k + 1 > 0$,

$$\rho < \frac{b}{q^6} - \frac{vB_R}{6q^6}.$$

When $B_R \geq 0$, this implies $\rho < |b|/q^6$. If $B \leq -1$, then $Wt < -L$ and (4), with positive κ, σ and (8), gives $n < 0$.

The integer exceptions are exactly $R \leq -4$ and $R = 0$. Write $P_{\text{mask}} = (M_C + qM_F)J > 0$. The exact outer identity is

$$R = (hC + q\alpha)(q^2 - 1) + P_{\text{mask}}. \tag{9}$$

For $R = -r \leq -4$, it implies $C < 0$ and

$$|C| > \frac{r}{h(q^2 - 1)} > \frac{r}{q^3}.$$

Thus $B = -(K_0 + w)|C| - F \le -1 - w|C|$ and $Wt < -(1 + w|C|)L$. The adverse correction is bounded by

$$\delta_0 = \frac{v(r-1)(r+1)(r-3)}{6q^6} < \frac{ws^2r^3}{6q^3}, \tag{10}$$

because $v = wq^3s^2$. Here $b = cT + r > r, c$, so $V > (F_* - 1)b$ and either choice of L satisfies $L > (F_* - 1)r^2$. Also $\Delta > Y^2 \ge s^2$ and $F_* - 1 = 4a(1 + a) > s^2$. Consequently $w|C|L > \delta_0$. Substitution gives

$$n < L/2 - (1 + w|C|)L + |b|/q^6 + \delta_0 < 0.$$

For $R = 0$, (9) again gives $C < 0$, while $\delta_0 = v/(2q^6) = ws^2/(2q^3)$. Now $b = cT \ge c$ and $V \ge (F_* - 1)b$, so $L > F_* - 1$ and $w|C|L > w(F_* - 1) > \delta_0$. The same contradiction follows. This handles every integer rank, including the cases where the new cubic correction has the unfavorable sign.

Therefore every positive zero has $B \ge 0$, which forces $C > 0$ because $F > 0$. The retained mask bound now applies:

$$3q < R < q^4 - 2. \tag{11}$$

Equation (7) gives $c > 2R$, whence $b, V > 0$. The retained normalized rank and signed auxiliary-index theorems have all their original hypotheses: $c^2 \mid I$, $c > 2R, c > 2p$, $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$. They give $p = R$. In particular R is positive odd; the unchanged mask parity gives even q and odd J.

2.5 The second-order index lemma

The new argument is elementary and works at every prime, including the primes dividing the displayed denominator 6.

Lemma 2.2 (Corrected index recovery)**.** *At every positive zero of* (3)*,*

$$n_0 = (R + 1)/2, \qquad q^6 \mid \bar{e}, \qquad \rho \in \mathbb{Z}_{\ge 0}, \qquad q^2 \mid s.$$

The conclusions hold as well for the square-transport control.

Proof. The Lucas polynomial has the finite expansion

$$\psi_{1+2v}(m) = \sum_{j=0}^{m-1} (4v)^j \binom{m + j}{2j + 1}. \tag{12}$$

For a direct derivation, its recurrence gives the formal generating function

$$\sum_{m \ge 1} \psi_{1+2v}(m) t^{m-1} = \frac{1}{(1 - t)^2 - 4vt} = \sum_{j \ge 0} \frac{(4v)^j t^j}{(1 - t)^{2j+2}}.$$

Coefficient comparison yields (12). In particular, for an explicitly determined nonnegative integer Z_m,

$$2\psi_{1+2v}(m) = 2m + 8v\binom{m + 1}{3} + 32v^2 Z_m. \tag{13}$$

The formula also covers $m = 1, 2$, when $Z_m = 0$.

Put $r_0 = (R + 1)/2$, an integer by the preceding rank argument. Then $B_R = 48\binom{r_0+1}{3}$, so $\bar{e} = e_6/6$ is an integer and

$$\bar{e} = k - 2r_0 - 8v\binom{r_0 + 1}{3}. \tag{14}$$

The integrality of z and the carrier give $6q^6 \mid hH(e_6 - 6q^4 s)$. Cancelling the literal common factor 6, now that (14) is integral, gives $q^6 \mid hH(\bar{e} - q^4 s)$. Because $h = q - 1$ and $H = 4Y(X+1) - 1 \equiv -1 \pmod{q}$, one has $\gcd(q, hH) = 1$. Hence

$$q^6 \mid \bar{e} - q^4 s, \qquad q^4 \mid \bar{e}. \tag{15}$$

No inverse of 6 modulo q has been taken.

Set $d = 2n_0 - R - 1 = 2(n_0 - r_0)$. Reducing (13) and (14) modulo $4v$ gives $\bar{e} \equiv d \pmod{4v}$. Since $q^3 \mid v$, equation (15) first gives $q^3 \mid d$. The exact Taylor difference is

$$6(\bar{e} - d) = vd\big[(2n_0)^2 + (2n_0)(R+1) + (R+1)^2 - 4\big] + 192v^2 Z_{n_0}. \tag{16}$$

Both terms are divisible by q^6. If a prime $\ell \mid q$ has $a = v_\ell(q) \geq 1$, then $v_\ell(6) \leq 1$, so

$$v_\ell(\bar{e} - d) \geq 6a - 1 \geq 4a.$$

Thus $q^4 \mid \bar{e} - d$, uniformly at $\ell = 2, 3$ as well. Together with (15), this gives $q^4 \mid d$.

The first base $2XY^2 + 1$ exceeds A. Were $n_0 \geq R$, monotonicity in the positive base and index would give $k/2 \geq \psi_A(R) = c$, contrary to $c > k$. Therefore $1 \leq n_0 < R$, and (11) gives $|d| < R < q^4$. Hence $d = 0$.

Only now return to the full expansion: equations (13)–(14) give $\bar{e} = 32v^2 Z_{r_0}$. Because $q^3 \mid v$, this proves $q^6 \mid \bar{e}$ and $\rho \geq 0$ integral. Finally, (15) gives $q^6 \mid q^4 s$, hence $q^2 \mid s$. □

The two congruence steps are essential. The first establishes divisibility of the index error by q^3; this makes the correction error small in the relevant prime valuations and upgrades it to the q^4 modulus needed for the ordinary size bound. The full q^6 denominator is recovered after the index has been identified.

2.6 Projection and restoration of the native source

Lemma 2.2 makes $\sigma = s/q^2$ integral and restores $q^3 \mid Y = qs$ before projection. The carrier now gives $\kappa - WB/h \in \mathbb{Z}$, so $H \mid hK$. Write $R = 2r + 1$. By (11), $r \geq 48$. The retained ratio theorem, uniformly valid at scale one, gives

$$Y > X^r/3, \qquad X^{r+1} < 3Y(X+1) < H.$$

The universal Pell projector and the retained entire-product estimate give

$$H \mid h(X - 2^R)(2^R X + 1), \qquad |h(X - 2^R)(2^R X + 1)| < H.$$

The product is zero, so $X = 2^R$ and κ is integral. No possibly nonunit factor of the product has been cancelled modulo H. The remaining carrier gives $h \mid WB$, and $W \equiv 1 \pmod{h}$ gives $h \mid B$. Thus the native quotient $t = B/h$ is a nonnegative integer.

All inherited fixed-minus parity and exact half-binomial rounding hypotheses have now been restored. The fresh reconstruction uses the original source base $Y(X+1) + 2$, which is distinct from the auxiliary base A used here. It produces an actual zero of the calibrated native source. The whole fixed coefficient tuple's retained native/refutation theorem therefore yields a finite syntactic ZFC refutation from every positive zero of (3).

2.7 Construction of positive witnesses

Conversely, a finite refutation supplies the parent's canonical outer data: $q^3 \mid Y$, $X = 2^R$, $R \equiv 3 \pmod{4}$, $R > 3q$ and integral $t = B/h \geq 0$, with the exact half-binomial interval. Define the new positive input

$s = Y/q$. Then $\sigma = s/q^2 = Y/q^3$ is an integer and at most Y. Using the scale-one bases, put

$$c = \psi_A(R), \qquad k = 2\psi_{2XY^2+1}((R+1)/2),$$
$$\tau = \chi_{2XY^2+1}((R+1)/2), \qquad \eta = c - kY, \quad \zeta = k - \eta.$$

The inherited uniform ratio estimate gives $\eta, \zeta > 0$. The full Lucas expansion gives $\rho \geq 0$ integral; projection gives positive integral $\kappa < Xc$.

A positive multiple of a finite Pell matrix period modulo $c^2\tau$ supplies one seed pair (f_0, I_0) with $c^2\tau \mid I_0$. Set $i = I_0/(c^2\tau) > 0$ and double this pair exactly as in the parent. It yields (6), $G = (\tau f_0)^2$ and the canonical positive auxiliary pair V, y, with the positive integral unshifted input

$$T = \frac{V + c + Rf^2}{cf^2}.$$

No private strong pair is reselected after its auxiliary congruences are specified. Moreover

$$f_0 > (A-1)I_0 > 2Xc, \qquad \kappa < Xc, \qquad \sigma \leq Y < Xc,$$

so $g = \tau f_0 > \kappa + \sigma$. Now set

$$n = gy + (1 + hL)t + \rho - \kappa - \sigma.$$

This is a strictly positive integer, and the resulting carrier is $z = gy$. The polynomial therefore vanishes. Every one of the ten independent inputs has been constructed positively. This completes both existential directions in Theorem 2.1.

2.8 Exact degree and validation

The relevant global degrees are

expression	Y	c	A	Δ	v	E	U	a
deg	2	3	4	8	6	14	7	36

expression	F_*	Q	G	b	V	M	Z $(L = bV)$
deg	72	80	50	4	76	11	93

The correction $v(R+1)(R-1)(R+3)$ has degree 18, below the leading transport part. For $L = bV$, the term $-QM^2GV^2$ uniquely dominates the final expression and has degree 304. For $L = V^2$, $\deg Z = 165$ and QZ^2 uniquely supplies degree 410.

At the smaller scale, cT and R both have degree four; the leading part of $b = cT - R$ must retain both. Put

$$q_0 = \beta J, \qquad k_0 = \eta + \zeta, \qquad C_0 = q_0 - F - \alpha, \qquad B_0 = k_0 sT - q_0^2(q_0 - F).$$

The entire highest homogeneous polynomials are

$$[P_{bV}]_{304} = -9 \cdot 2^{12} q_0^{128} w^{34} s^{78} i^{14} k_0^{44} B_0^2,$$
$$[P_{V^2}]_{410} = 9 \cdot 2^{16} q_0^{166} w^{46} s^{104} i^{20} k_0^{60} B_0^4 C_0^2.$$

For example, their respective coefficients at

$$J^{128} w^{34} s^{80} i^{14} \eta^{46} T^2, \qquad J^{166} w^{46} s^{108} i^{20} \eta^{64} T^4 F^2$$

are $-9 \cdot 2^{12}\beta^{128}$ and $9 \cdot 2^{16}\beta^{166}$, both nonzero at the authentic fixed tuple. These give global degree certificates, rather than sampled line degrees.

The independent exact checker binds all actual evaluator rows through 37 free-polynomial cuts per circuit. It recounts the live gates, checks every fixed coefficient byte, and propagates the complete highest

homogeneous forms: 274 terms for the principal equation and 3,750 for the control. Actual operator corruption checks reject 175 modified principal artifacts and 173 control artifacts. Normal and optimized Python runs agree. A separate retained reconstruction script regenerates all four fixed numerals from their finite recipe and checks them against the stored bytes. The Taylor identity (16) is checked exactly; separate finite Lucas recurrence controls are explicitly diagnostic.

Those finite checks certify the equations and algebraic certificates. They do not replace the unbounded arithmetic proof above or the named inherited calibration, and do not decide the existence of a ZFC refutation. The new work has also received a separate ordinary mathematical review within this AI-assisted workflow; that is not an outside human referee's endorsement.

3 A direct quartic from one certificate for two front signs

The second contribution is a local arithmetic simplification of the direct C222 construction. Its principal equation has 248 unrestricted integer variables, degree four, 5,321 collected monomials, and coefficient height 128,446. It is the sum of the squares of the 182 explicit quadratic residuals in `equations/integer248/equation.json`. Here and below, coefficient height is measured after collecting the entire polynomial.

Profile	Variables	Residuals	Degree	Terms	Height
Common Remainders parent	252	182	4	5,340	128,446
Principal	248	182	4	5,321	128,446
Balanced projection	246	180	4	5,048	195,955
Dimension projection	245	179	4	21,845	314,701

These are three achieved projections of one new raw source, not an assertion that every better projection has been excluded. The principal profile strictly improves the dimension and term count of its matched parent while preserving degree, residual count, and coefficient height. The new mathematical step is a proof that one product certificate can replace two front sign certificates, even though the product alone has an unwanted negative branch. The exact factorial digit already present in the source excludes that branch.

3.1 Literal source change and inherited interface

Put

$$G(a,b,c) = a^2 + b^2 + c(c+1).$$

This is nonnegative on $\mathbb{Z}^3$ and represents every nonnegative integer. Indeed, the three-square theorem represents $4n+1$ as three squares; its residues modulo four put exactly one odd square in that representation. Writing it as $(2c+1)^2$ and the others as $(2a)^2, (2b)^2$ gives $n = G(a,b,c)$.

Use the following semantic names for the unchanged source coordinates:

$$r = \texttt{fact0.remainder}, \quad s = \texttt{fact0.slack}, \quad F = \texttt{fact0.output},$$
$$B = \texttt{choose0.output}, \quad V = \texttt{pow0.output}, \quad q = \texttt{pow2.output}.$$

The old front separately imposed $r \geq 0$, $s \geq 0$, and $o \geq 0$, where $q = B + o + 1$. Replace its two first ternary guards by

$$(r+1)(s+1) = 1 + G(a,b,c). \tag{17}$$

This is the quadratic row $rs + r + s - a^2 - b^2 - c^2 - c = 0$; it requires no multiplication gate. Compose the previously supplied odd-square radix restriction by deleting $q = B + o + 1$ and its private guard, and

inserting

$$q = (2z + 1)^2. \tag{18}$$

The two remaining guards—the strict squared Pell cap and the common normalized remainder norm—are unchanged. All other source rows, all eighteen power requests, and all endpoint and Horn-flow relations are retained. The old coordinate o occurs only in its deleted bound row and can therefore be restored freely at the end.

The starting point is the direct component of `ConZFC_Common_Remainders_Radix47_2026-10-25.zip`, at repository snapshot `3ecd6a6cf53d47519228328567f45fb7ee9781d9`. Its `NORMALIZED_REMAINDERS_PROOF.md`, together with the retained `ORIENTATION_FIRST_PROOF.md` and `STRICT_BUDGET_PROOF.md`, supplies the following interface *before* the removed front signs are used. An integer $W \geq 258$ and a signed Pell center A satisfy

$$|A| > W^W, \qquad |n|, |k|, |m| < W \tag{19}$$

for every requested base, index, and output. The corresponding principal Pell coordinates are canonical at the signed nominal index k. The frame uses squares of the proposed atoms, and the squared semantic anchor with bounded pooled errors orients both Pell coordinates. Consequently those conclusions do not presuppose the signs of F, B, q or of the independent CRT representatives. We retain that inherited orientation theorem and extend its subsequent power argument to signed bases explicitly below.

3.2 The signed-base extension

Lemma 3.1 (Signed bases)**.** *Under the inherited interface* (19)*, a power request with* $2 \leq |n| < W$ *forces* $k \geq 0$ *and* $m = n^k$*. The assertion holds for either sign of* n *and either sign of* A*.*

Proof. Set $T = 2An - n^2 - 1$. The actual output row is

$$m \equiv X_k(A) + (n - A)Y_k(A) \pmod{T}.$$

The canonical signed pairs have $X_{-j}(A) = X_j(A)$ and $Y_{-j}(A) = -Y_j(A)$. Furthermore

$$|T| \geq 2|A||n| - n^2 - 1 \geq 4|A| - (W - 1)^2 - 1 > W^W + W,$$

where the last inequality follows from $W \geq 258$ and $|A| > W^W$. The absolute value on A is essential: no positive-center premise has been introduced. Since $T \equiv -1 \pmod{n}$, n is invertible modulo T.

Substituting $n - A$ into the Pell identity gives $(n - A)^2 \equiv A^2 - 1 \pmod{T}$. The canonical Pell addition identities therefore give $m \equiv n^k \pmod{T}$ when $k \geq 0$. In that case

$$|m - n^k| \leq |m| + |n|^k < W + W^W < |T|,$$

so the congruence is an equality. If $k = -j < 0$, the signed pair is the inverse of the index-j Pell unit; its norm is one. Thus $mn^j \equiv 1 \pmod{T}$. Now $1 \leq j \leq W - 1$ and $|m| \leq W - 1$, so

$$|mn^j - 1| \leq (W - 1)^W + 1 < |T|.$$

It follows that $mn^j = 1$, impossible in $\mathbb{Z}$ because $|n|^j \geq 2$. Hence $k \geq 0$ and the first argument applies. □

For the separate base-one case, a known nonnegative index gives $m \equiv 1 \pmod{2A - 2}$; the same size argument gives $m = 1$. This case is needed because $F = 0$ has not yet been excluded. The known index $t \geq 16$ in the request at base $F^2 + 1$ consequently gives

$$V = (F^2 + 1)^t \geq 1 \tag{20}$$

for every initially possible integer F.

3.3 Both front branches have the needed size bound

The untouched ceiling relations are

$$r + s = B - 1, \qquad BF = V + s. \tag{21}$$

The right side of (17) is positive. Since its two factors are integers, it leaves precisely the alternatives

$$\text{(I) } r, s \geq 0, \qquad \text{(II) } r, s \leq -2.$$

In case I, $B \geq 1$ and $BF = V + s \geq 1$ force $F \geq 1$. Since $V = (F^2 + 1)^t > F^2$, the inequality $V \leq BF$ then gives $B > F$. In particular $B \geq 2$ and

$$V \leq BF < B^2.$$

In case II put $B = -b$, so $b \geq 3$. If $F = 0$, the base-one case of (20) gives $V = 1$, and $BF = V + s$ would give $s = -1$. If $F \geq 1$, combining (21) gives

$$r = (1 - F)B + V - 1 > 0,$$

because $(1 - F)B \geq 0$ and $V > 1$. Both possibilities contradict case II. Therefore $F = -f$ for some $f \geq 1$. The same two ceiling identities now give

$$bf + 2 \leq V \leq (f + 1)b - 1 < (f + 1)b. \tag{22}$$

For $f \geq 1$ and $t \geq 16$, $(f^2 + 1)^t > (f + 1)^2$. Comparing this with (22) shows $b > f + 1$, and hence $V < b^2$.

Thus, in both branches, defining $f = |F|$ and $b = |B|$ gives

$$f \geq 1, \qquad b \geq 2, \qquad V = (f^2 + 1)^t < b^2. \tag{23}$$

This deduction has used no factorial fixed point or binomial-digit interpretation. That separation is what makes the later sign recovery noncircular.

3.4 Recovering the radix before resolving the branch

Equation (18) first gives $q \geq 1$, so $q + 1 \geq 2$. Each of the nine actual CRT representatives a_i is an exponent in a retained request at base $q + 1$. Lemma 3.1 therefore makes each $a_i \geq 0$ and authenticates its output. Set

$$H_0 = 1 + \sum_{i=1}^{9} a_i \geq 1.$$

These are actual representatives; several delivered coordinates are shifted by one and need not themselves be nonnegative.

Apply the lemma next to the retained request $N = B^x$. Its base has $|B| \geq 2$ by (23), so it forces $x \geq 0$ even while $q = 1$ is still an initial possibility. The unchanged cofinal beta choice is

$$y = V(V - 1) > f + 2.$$

In unshifted semantic coordinates, the retained endpoints are

$$x = F + 1 + (y + 1)u_0, \qquad x = 1 + \big(1 + (t + 1)y\big)u_t. \tag{24}$$

The second equality excludes $x = 0$ because its bracket is an integer greater than one. If $x = 1$, the first would say $F + (y + 1)u_0 = 0$, impossible since $0 < |F| < y + 1$. Consequently $x \geq 2$, and the second equality then forces $u_t \geq 1$. Thus

$$x \geq 2 + (t + 1)y \geq 8. \tag{25}$$

The coordinate translations used in the emitted polynomial do not alter these semantic identities; their explicit shifts are recorded separately.

The request $q = (BH_0)^x$ now has $|BH_0| \geq 2$ and is authenticated by the lemma. Since its output is positive, $B < 0$ would force x even. Therefore in either branch

$$N = b^x > 0, \qquad q = b^x H_0^x \geq b^x, \qquad N \mid q. \tag{26}$$

In particular $q > 1$. All denominator requests are now genuine positive powers of q. Write

$$M = q^x, \qquad D_0 = q^t, \qquad D_1 = q^{t+1}, \qquad D_y = q^{y+1}.$$

By (25), these three denominators divide M. Nothing in this argument has assumed $F = t!$ or the sign of B.

3.5 The exact factorial digit chooses the positive branch

For $m \geq 1$, $t \geq 1$ and $0 \leq j \leq t$, the elementary estimate

$$\binom{m + t}{j} \leq (m + 1)^t \tag{27}$$

holds. One proof is induction on t: Pascal's identity bounds each nonedge entry in the next row by $2(m + 1)^t \leq (m + 1)^{t+1}$; the new edge is bounded by $\prod_{i=1}^{t+1}(1 + m/i) \leq (m + 1)^{t+1}$. Apply this with $m = F^2 \geq 1$. Every coefficient through position t of $(q+1)^{F^2+t}$ is at most $V < b^2$. Its true low remainder at position t is therefore

$$r^0 = \sum_{j=0}^{t-1} \binom{F^2 + t}{j} q^j, \qquad 0 \leq r^0 < \frac{VD_0}{q - 1} < \frac{D_0}{32}.$$

For the last inequality, (26) and (25) give

$$\frac{V}{q - 1} < \frac{b^2}{q - 1} < \frac{2}{b^{x-2}} \leq \frac{1}{32}.$$

Thus the displayed sum really is the ordinary remainder modulo D_0; the polynomial tail is divisible by D_0.

Here is the precise role of the inherited common norm. Its normalized digit rows have $D_i v_i = M$ and $v_i U_i = MQ_i + R_i$. Since $D_i, M > 0$, v_i is a positive integer. The digit row makes $v_i \mid R_i$, so $r_i = R_i/v_i$ is an integer. The norm equation

$$M^2 = 1 + 4\sum_{i=1}^{11} R_i^2 + G(\alpha, \beta, \gamma) \tag{28}$$

gives $|R_i| < M/2$, hence $|r_i| < D_i/2$. For the factorial row, $r_i \equiv r^0 \pmod{D_0}$. Together with $0 \leq r^0 < D_0/32$, this forces $r_i = r^0$. No sign of B is used in this recovery.

The exact factorial digit row is now

$$(q + 1)^{F^2+t} = D_0(B + qh) + r^0.$$

Divide by D_0 after subtracting r^0 and reduce modulo q to obtain

$$B \equiv C \pmod{q}, \qquad C = \binom{F^2+t}{t} > 0. \tag{29}$$

Unlike the old proof, this argument does not yet assume $0 \le B < q$. Instead, (23) and the coefficient estimate give

$$|B - C| \le b + C \le b + V < b + b^2 < q,$$

the last inequality following from $q \ge b^x$, $b \ge 2$ and $x \ge 8$. Equation (29) therefore gives $B = C > 0$. Case II is impossible. We have recovered $r, s \ge 0$, $F \ge 1$, and $q > B$. Restore the old private slack by $o = q - B - 1 \ge 0$ and fill all three deleted ternary guards using surjectivity of G.

At this point the previous Common Remainders source has an integer solution. Its exact factorial fixed-point theorem gives $F = t!$. Its remaining ten low-digit estimates, prime-power CRT range decoding, and Horn-flow interpretation therefore apply with their original premises. The new argument has restored those premises in the order needed: size bound, radix, normalized remainder, factorial digit, front signs, then factoriality.

3.6 Completeness and the parity restriction

For the converse, begin with a finite C222 refutation and use the retained cofinal arithmetic construction. The odd-square radix restriction is inherited from the separate Centered Residues proof, included as `INHERITED_ODD_SQUARE_PROOF.md`; its compatibility with the present construction can be seen explicitly.

After factoriality, $F = t!$ is even. Each $i \le t$ divides F and F^2/i is even, so

$$B = \binom{F^2+t}{t} = \prod_{i=1}^{t} \left(1 + \frac{F^2}{i}\right)$$

is odd. Also $y = V(V-1)$ is even, so every beta CRT modulus is odd. The CRT progression for x consequently has odd period, allowing x to be chosen even and as large as required. Now $N = B^x$ and N^2 are odd. The independent amplified CRT representatives can be shifted by multiples of N^2 without changing their required residues. Adjusting a third representative's parity makes H_0 odd while leaving the two representatives used for the dominant anchor's parity free to retain opposite parities. The dominant lift can still be made large enough for all strict bounds. Thus BH_0 is odd and x is even; $q = (BH_0)^x$ is the square of an odd integer and (18) has an integer witness z.

The genuine old front has $r, s \ge 0$, so $rs + r + s \ge 0$ has a G representation and satisfies (17). The old cofinal low-remainder estimates give $0 \le r_i^0 < D_i/32$ for all eleven digits. Setting $R_i = (M/D_i)r_i^0$ gives $R_i < M/32$ and

$$4\sum_{i=1}^{11} R_i^2 < \frac{44}{1024} M^2 < M^2.$$

Hence the nonnegative integer $M^2 - 1 - 4\sum_i R_i^2$ has a G representation, preserving (28). Every other retained row is the old row with its cofinal witness. This constructs a zero of the new raw source.

The raw modification therefore preserves *existence* of integer zeros in both directions. It is not a bijection on all arbitrary old private-witness tuples. The later monic projections and integral translations, in contrast, have explicit reverse-restoration maps.

3.7 Exact endpoint and the logical calibration chain

Theorem 3.2. *Let $P_{248} \in \mathbb{Z}[x_0, \ldots, x_{247}]$ be the sum of the squares of the 182 delivered residuals in the principal profile. With the retained direct-C222 arithmetic and logical calibration,*

$$\mathrm{ZFC} \vdash \Big(\exists x \in \mathbb{Z}^{248}\ P_{248}(x) = 0\Big) \longleftrightarrow \textit{there exists a finite ZFC refutation.}$$

Consequently ZFC proves that the absence of integer zeros of P_{248} is equivalent to Con(ZFC)*. The same conclusions hold for the two delivered 246- and 245-variable quartics.*

Proof. Over the integers a finite sum of squares vanishes exactly when every residual vanishes. The supplied integral translations are bijections; each monic elimination has an integer polynomial restoration, replayed in reverse order. The independently checked Horn factoring identities preserve the source's solution existence. The preceding soundness and completeness arguments identify zeros of the new source with zeros of the Common Remainders source. The retained arithmetic proof decodes those zeros into finite C222 refutations, and constructs arithmetic witnesses from such refutations.

The last two logical bridges are the identified repository theorems `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Thus the complete chain is: explicit quartic zero, new raw arithmetic witness, old Common Remainders witness, finite C222 refutation, finite ZF refutation, and finite ZFC refutation. Every arrow used here is either the new arithmetic argument above, a finite exact algebraic certificate, or the named inherited calibration. The first-order arithmetical statement that no finite syntactic ZFC refutation exists is Con(ZFC); negating the existence equivalence gives the stated result. □

The package retains all 88 files in the parent's arithmetic and proof closure, pinned byte-for-byte. Its new finite verifier imports the retained independent algebra checker, not the producing eliminator. It reconstructs the complete raw mutation, checks 18 Horn-factor edits, replays every monic restoration and translation, and expands each full sum of squares. The principal profile has 203 monic elimination steps and 70 translated coordinates. Independent runs under ordinary Python and under `python -O` pass, including five rejected deliberate mutations; a clean rebuild reproduces all 22 generated JSON files byte-for-byte. These are algebraic checks of the emitted artifacts. The unbounded arithmetic proof remains an ordinary mathematical proof, with a separate independent review; no new end-to-end Lean elaboration or proof of consistency is being claimed.

4 Trading degree for fewer direct proof coordinates

The preceding sign bootstrap gives a direct equation in 248 unrestricted integers. We now reduce its dimension by allowing higher degree. This route retains the direct C222 proof calculus: no K215 simulation, Jones universal equation, or large fixed coefficient tuple is introduced. The elimination principle is standard. The contribution here is the complete family of explicit polynomials and the certified tradeoffs below.

Write the preceding source as

$$P_{248}(x) = \sum_{j=0}^{181} R_j(x)^2, \qquad x = (x_0, \ldots, x_{247}) \in \mathbb{Z}^{248}.$$

The complete quadratic residual list is retained without change in `source/equation248.json`; its source hash is recorded in `SOURCE.json`. The pivot tables in Appendix C, together with that literal residual list, specify every new polynomial by finite substitution.

Theorem 4.1 (Acyclic monic elimination)**.** *Let $R_0, \dots, R_{m-1} \in \mathbb{Z}[x_0, \dots, x_{N-1}]$. Select different rows and different coordinates so that each selected row has the form*

$$R_{r(v)} = \varepsilon_v x_v + f_v, \qquad \varepsilon_v \in \{-1, 1\}, \qquad x_v \text{ does not occur in } f_v.$$

Suppose the directed graph on selected coordinates, with an edge $v \to w$ when x_w occurs in f_v, is acyclic. List the retained source indices as $i_0 < \cdots < i_{n-1}$, and let $y = (y_0, \dots, y_{n-1})$. Define integer polynomials $F_v(y)$ by $F_{i_j} = y_j$ on retained coordinates, and recursively by

$$F_v = -\varepsilon_v f_v(F)$$

on selected coordinates. If I is the set of unselected rows, put

$$Q(y) = \sum_{j \in I} R_j(F(y))^2.$$

Projection to retained coordinates is a bijection between the integer zero sets of $\sum_j R_j^2$ and Q. Its inverse is the polynomial map $y \mapsto (F_0(y), \dots, F_{N-1}(y))$.

Proof. At an integer zero of the source sum of squares, every residual is zero. Each selected row therefore forces exactly $x_v = -\varepsilon_v f_v$. Induction along a reverse topological order of the acyclic graph gives $x_v = F_v(y)$ for every coordinate. The unselected residuals vanish, so $Q(y) = 0$.

Conversely, if $Q(y) = 0$, every remaining squared integer residual is zero. Set $x = F(y)$. The selected residuals vanish by their definitions, and the unselected ones by $Q(y) = 0$. Thus the source polynomial is zero. Every restored coordinate is an integer because all the substitutions have integer coefficients. The same induction proves uniqueness. No division, nonzero denominator, congruence, or sign choice is needed. □

The unrestricted integer domain matters. Applying this theorem to a nonnegative-domain equation would require additional proofs that all restored coordinates remain nonnegative at every zero. We use the integer source throughout.

4.1 The resulting explicit family

Every row in the following table defines one ordinary polynomial. Its complete arithmetic DAG uses only integer constants, its stated input variables, and binary addition, subtraction and multiplication. Intermediate registers abbreviate polynomials and are not additional quantified unknowns. Degree is exact. The last column is a proved bound on the height after collecting all coefficients; it is not the largest constant appearing in the DAG.

Integer unknowns	Remaining residuals	Degree	Height bound
225	159	6	$< 2^{27}$
153	87	8	$< 2^{47}$
144	78	12	$< 2^{50}$
125	59	16	$< 2^{80}$
117	51	32	$< 2^{144}$
111	45	64	$< 2^{173}$
102	36	256	$< 2^{691}$
94	28	65,540	$< 2^{164870}$

The degree-16 equation removes 123 of the source's 248 coordinates, or 49.6%. The degree-256 equation removes 146, or 58.9%. These are alternatives within the direct proof-calculus route, not global dimension records among all coefficient encodings. The 94-variable equation illustrates the cost of prioritizing dimension alone: its degree and coefficient bound are much larger.

For the 153-variable octic the package also supplies every collected coefficient. Its exact expansion has 188,579 nonzero monomials and height

$$750{,}273{,}658{,}864.$$

Its 87 fully substituted residuals contain 3,476 monomials in total. This exact height is stronger than the general 2^{47} bound. The expansion makes the support cost of eliminating coordinates explicit.

4.2 Exact degree and coefficient bounds

Give each retained coordinate degree one. For each restored coordinate propagate the maximum, over its monomials, of the sum of the previously assigned coordinate degrees. Apply the same rule to each remaining residual. If the largest bound is d, then $\deg Q \le 2d$.

To prove equality, the supplied certificate gives a remaining residual of degree bound d whose degree-d homogeneous part is nonzero when the retained coordinate with source index i is set to $i+3$, reduced modulo 1,000,000,007. A nonzero modular value proves a nonzero integer value; no primality assumption is needed. At this integer point the sum of the squares of all degree-d homogeneous parts is positive. It follows that $\deg Q = 2d$. This is an exact nonvanishing certificate, not probabilistic polynomial identity testing.

Lemma 4.2 (A propagated height bound)**.** *Set $L_v = 1$ for each retained coordinate. For a restored polynomial $F_v = \sum_M c_M \prod_{j\in M} F_j$, set*

$$L_v = \sum_M |c_M| \prod_{j\in M} L_j.$$

Use the same rule to obtain a bound L'_j for each remaining residual. Then every coefficient of Q has absolute value at most

$$B_Q = \sum_{j\in I} (L'_j)^2.$$

If this positive integer has bit length b, every coefficient has absolute value strictly below 2^b.

Proof. For an integer polynomial f, let $\|f\|_1$ be the sum of the absolute values of its collected coefficients. The triangle and convolution inequalities give

$$\|f+g\|_1 \le \|f\|_1 + \|g\|_1, \qquad \|fg\|_1 \le \|f\|_1 \|g\|_1.$$

Induction through the substitutions proves $\|F_v\|_1 \le L_v$. Applying the same inequalities to the residuals and their squared sum proves $\|Q\|_1 \le B_Q$. Each coefficient is bounded by this norm, and $B_Q < 2^b$ by the definition of bit length. □

All stated bit bounds are obtained by exact integer arithmetic. The largest literal constant in these DAGs is 256; that different statistic does not replace the expanded coefficient bounds in the table.

4.3 Verification and the consistency theorem

The independent checker expands each circuit block as a polynomial in the coordinates already available. These local expansions have degree at most two, even when the fully substituted polynomial has degree 65,540. It checks the blocks coefficient by coefficient against their source rows, verifies the complete row partition and acyclic restoration, and checks the final sum of squares. A separate sparse engine expands the octic completely. Exact nonzero partial derivatives modulo 1,000,000,007 certify that every declared input occurs in each collected polynomial. These tests accompany the ordinary proof; they do not replace its unbounded source calibration.

By Theorem 4.1, each displayed Q has an integer zero if and only if P_{248} has one. Compose this elementary, ZFC-formalizable bijection with the already proved existence equivalence for the fused front-sign source and its retained C222/ZFC calibration. For every row of the table this gives

$$\mathrm{ZFC} \vdash [Q \text{ has no zero in its stated integer domain}] \iff \mathrm{Con(ZFC)}.$$

This establishes an equivalence, without assuming or proving either side. The new finite algebra is not claimed to be a fresh end-to-end proof-assistant development.

5 Verification, reproduction, and use of the archive

The archive is designed for local reproduction without a network connection. Its mathematical data and checkers use Python's standard library. The PDF can be rebuilt with an ordinary pdfLaTeX installation. From the package root:

```
python3 -B verify.py
python3 -B -O verify.py
python3 -B verify.py --rebuild
python3 build_report.py
```

The first command checks the sealed file manifest and component/source bindings, then runs the independent finite audits. The second also disables Python assertions throughout the component checks; no substantive check is allowed to disappear with optimization. The third additionally reconstructs the delivered equation files and compares their bytes with the release. The report builder updates the PDF and its single-file TeX edition. Such a rebuild intentionally changes a sealed document if the local TeX engine differs, so a release's manifest should be checked before rebuilding it.

For the compact pair the separate checker binds the actual circuit gates to 37 exact polynomial cuts per equation, propagates the entire highest homogeneous form, and verifies five additional algebraic identities. It rejects 175 actual gate mutations for the principal and 173 for its control. The four fixed numerals are reconstructed from the complete seed recipe, and every delivered coefficient payload is authenticated. The finite recurrence and index tests are diagnostic controls in addition to the unbounded proof, not evidence that a consistency statement has been decided by finite search.

For the direct quartics the checker independently reconstructs the changed source, checks the 18 Horn factoring identities, all monic projections, all integral translations, the entire collected quartics, and the unchanged parent file pins. The principal has 203 monic projection steps and 70 translated coordinates. A clean rebuild reproduces all 22 generated JSON artifacts for the three profiles. Five deliberately corrupted new artifacts are rejected; the parent's own audit is retained and run as well.

For the eight higher-degree forms the checker locally expands every circuit block in previously defined source coordinates. Every source row is accounted for exactly once, either as a solved monic definition or as a remaining residual. This verifies the entire polynomial substitution without expanding the

highest-degree final forms. Separate nonzero modular derivative witnesses prove that every listed input actually occurs in the resulting polynomial. The 153-variable octic is also expanded in full by a separate sparse-polynomial engine. Seven deliberate corruptions test the algebraic certificate rejection path. The complete coefficient norm bound and a nonzero highest-degree witness establish the reported height bounds and exact degrees; they are not floating-point estimates.

5.1 Files that define the result

The authoritative compact circuit is `compact/equation_bV.json`. The principal direct equation is `direct_quartics/equations/integer248/equation.json`, together with its fully collected `expanded_polynomial.json`. The eight projected circuits are in `higher_degree/equations/`. Their source file is byte identical to the principal 248-variable equation; the top-level verifier checks this binding explicitly.

All 182 principal residuals are printed in Appendix B. Together with the recursive elimination rule and the complete pivot tables in Appendix C, they give a finite paper specification of every projected polynomial. The JSON circuits are convenient executable versions of those fixed definitions. The complete fixed coefficient recipe for the ten-positive form is printed in Appendix A.

Each component contains its proof, producer, independent verifier, exact evaluator or circuit format, source hashes and successful verification receipts. `review/` holds the additional scoped mathematical reviews and root-level replays. `SOURCE_LEDGER.json` records all inspected incoming archive hashes. `MANIFEST_SHA256.json` binds the final release files. Original licenses and notices are preserved.

6 What the new ideas change, and what remains open

The compact result demonstrates a useful design principle: an index error need not be encoded only to first order. When a Pell/Lucas congruence falls short of the modulus needed for a bounded rank, subtracting a short Taylor term can increase usable precision without adding an existential variable. Here the correction is cubic in the target rank, but its degree is far below the dominant transport term. The cost is a new signed-transport proof and careful treatment of primes dividing its denominator. The result succeeds because those costs can be paid without destroying the final degree gain.

The direct result illustrates a different saving. Two nonnegativity certificates can sometimes be replaced by one product certificate if an existing exact digit condition chooses the correct sign branch. The difficult part is the order of the proof: size bounds and power semantics must remain valid in the unwanted branch, before factoriality or the desired signs are known. The new signed-base argument supplies that order.

The higher-degree family uses standard monic substitution. Its contribution is the concrete, fully specified set of eliminations and certified tradeoff points on the new source. It should not be described as a new general elimination theorem. Compared with a quartic-only optimization, it exposes how many of the remaining unknowns are merely arithmetic intermediate values. Its large-degree endpoint is included as a measured boundary point, not selected as the most practical presentation.

Further reductions remain possible. A still smaller substitution such as $Y = q + s$ would change the crucial coprimality $H \equiv -1 \pmod{q}$ and the index-recovery divisibility; no proof for that substitution is supplied. Higher-order corrections also need to pay for their signed tails and their small-prime denominators. For the direct family, the delivered projection orders are achieved points, not a search-completeness certificate. More systematic degree/height-aware substitution or another semantic guard recovery could improve them. None of these prospects is counted as a completed result in the tables.

A Complete fixed coefficient recipe for the compact equation

This appendix reproduces the retained directly indexed BFS943 constant recipe. Its local symbols are scoped to this appendix: the integer called K here is K_0 in Section 2, and the radix B here is unrelated to the variable expression $B = (K_0 + w)C - F$ there. Every range is fixed. Thus this is a complete finite definition of four integer coefficients, not an extra quantified condition. The delivered binary coefficients are checked against this recipe, including every digit of the seed.

The native tuple is inherited, with its state/color conjugacy and calibration; it is not the new contribution of this paper. See `compact/parent/proof_sources/indexed_native.md` and the accompanying source ledger for the precise boundary.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ &\quad (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ &\quad (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j+2\ (j = 2q+b < 430)$	$((1-d_j)l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$432+b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434+b\ (b = 0, 1)$	$(0, b, 0, b)$
$424+2r_q+b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q,b))$
$424+2l_q+b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q,b))$
$924+j\ (0 \le j < 4)$	$(0, j+430\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497+t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$^{934+3a+b-6\lfloor a/4 \rfloor}_{a\in\{0,1,4\},\ b=0,1,2}$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1-m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j) D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

B All residuals of the principal direct quartic

The variables are exactly $x_0, \dots, x_{247} \in \mathbb{Z}$. Define

$$\boxed{P_{248}(x) = \sum_{j=0}^{181} R_j(x)^2 = 0.}$$

The following list is complete and ordered. There are no additional domain restrictions or unlisted rows. Its source file has SHA-256

`c51ebc12534db042bd1c7bf7c59f84602f51625e2d952dffa82fa054b5d92192`.

The variable-to-source correspondence and every restoration are delivered in the accompanying certificates. For the higher-degree forms, the row and coordinate numbers in this list are precisely those used by the pivot tables; no renumbering is implicit.

$$R_0 = -x_3^2 + x_4$$

$$R_1 = -1 - x_3x_5 + x_5 + x_6 - x_7$$

$$R_2 = x_0 - x_0x_2 - x_2 + x_{10}$$

$$R_3 = -1 + x_1 - x_2x_8 - x_3$$

$$R_4 = -1 + x_1 - x_9x_{10}$$

$$R_5 = 1 + x_4 - x_5x_{21} + x_{19}$$

$$R_6 = -1 + 2x_{11} - x_{11}^2 - 2x_{11}x_{12} + 2x_{12} - x_{12}^2 + x_{22}$$

$$\begin{aligned} R_7 = {} & -5 - 30x_{11} - 36x_{11}^2 - 96x_{11}x_{12} - 24x_{11}x_{22} - 40x_{12} - 64x_{12}^2 \\ & - 32x_{12}x_{22} - 10x_{22} - 4x_{22}^2 + x_{23} \end{aligned}$$

$$R_8 = -2 + 2x_{11} - x_{11}^2 - 4x_{11}x_{23} + 4x_{23} - 4x_{23}^2 + x_{24}$$

$$\begin{aligned} R_9 = {} & 1 + 6x_{11} - 36x_{11}^2 - 192x_{11}x_{23} - 24x_{11}x_{24} + 16x_{23} - 256x_{23}^2 \\ & - 64x_{23}x_{24} + 2x_{24} - 4x_{24}^2 + x_{25} \end{aligned}$$

$$R_{10} = -2 - x_{12}^2 - 2x_{12}x_{13} - x_{13}^2 + x_{26}$$

$$\begin{aligned} R_{11} = {} & -9 + 7x_{11} - x_{11}^2 - 4x_{11}x_{12} - 4x_{11}x_{26} + 12x_{12} - 4x_{12}^2 \\ & - 8x_{12}x_{26} + 12x_{26} - 4x_{26}^2 + x_{27} \end{aligned}$$

$$R_{12} = -1 + 2x_{11} - x_{11}^2 - 2x_{11}x_{13} + 2x_{13} - x_{13}^2 + x_{28}$$

$$\begin{aligned} R_{13} = {} & -6x_{11} - 144x_{11}^2 - 192x_{11}x_{12} - 192x_{11}x_{13} - 48x_{11}x_{22} \\ & - 48x_{11}x_{28} - 8x_{12} - 64x_{12}^2 - 128x_{12}x_{13} - 32x_{12}x_{22} - 32x_{12}x_{28} \\ & - 64x_{13}^2 - 32x_{13}x_{22} - 32x_{13}x_{28} - 2x_{22} - 4x_{22}^2 - 8x_{22}x_{28} \\ & - 4x_{28}^2 + x_{29} \end{aligned}$$

$$R_{14} = -4x_{27}^2 - 8x_{27}x_{29} - 4x_{29}^2 + x_{30}$$

$$\begin{aligned} R_{15} = {} & -6 + 16x_{13} - 16x_{13}^2 - 16x_{13}x_{26} - 16x_{13}x_{28} + 8x_{26} - 4x_{26}^2 \\ & - 8x_{26}x_{28} + 8x_{28} - 4x_{28}^2 + x_{31} \end{aligned}$$

$$\begin{aligned} R_{16} = {} & 1 + 6x_{11} - 36x_{11}^2 - 96x_{11}x_{12} - 48x_{11}x_{13} - 24x_{11}x_{22} - 48x_{11}x_{28} \\ & - 24x_{11}x_{31} + 8x_{12} - 64x_{12}^2 - 64x_{12}x_{13} - 32x_{12}x_{22} - 64x_{12}x_{28} \\ & - 32x_{12}x_{31} + 8x_{13} - 16x_{13}^2 - 16x_{13}x_{22} - 32x_{13}x_{28} - 16x_{13}x_{31} \\ & + 2x_{22} - 4x_{22}^2 - 16x_{22}x_{28} - 8x_{22}x_{31} + 8x_{28} - 16x_{28}^2 \\ & - 16x_{28}x_{31} + 4x_{31} - 4x_{31}^2 + x_{32} \end{aligned}$$

$$\begin{aligned} R_{17} = {} & -10 - 5x_{11} - x_{11}^2 - 12x_{11}x_{12} - 16x_{11}x_{13} - 4x_{11}x_{26} - 36x_{12} \\ & - 36x_{12}^2 - 96x_{12}x_{13} - 24x_{12}x_{26} - 48x_{13} - 64x_{13}^2 - 32x_{13}x_{26} \\ & - 12x_{26} - 4x_{26}^2 + x_{33} \end{aligned}$$

$$\begin{aligned} R_{18} = {} & -2 - 24x_{11} - 36x_{11}^2 - 12x_{11}x_{12} - 96x_{11}x_{13} - 24x_{11}x_{28} - 3x_{12} \\ & - x_{12}^2 - 16x_{12}x_{13} - 4x_{12}x_{28} - 32x_{13} - 64x_{13}^2 - 32x_{13}x_{28} \\ & - 8x_{28} - 4x_{28}^2 + x_{34} \end{aligned}$$

$$R_{19} = 2 - 4x_{33}^2 - 8x_{33}x_{34} - 4x_{34}^2 + x_{35}$$

$$\begin{aligned} R_{20} = {} & -12 + 16x_{11} - 4x_{11}^2 - 8x_{11}x_{13} - 8x_{11}x_{22} - 8x_{11}x_{26} + 16x_{13} \\ & - 4x_{13}^2 - 8x_{13}x_{22} - 8x_{13}x_{26} + 16x_{22} - 4x_{22}^2 - 8x_{22}x_{26} \\ & + 16x_{26} - 4x_{26}^2 + x_{36} \end{aligned}$$

$$\begin{aligned} R_{21} = {} & -3 - 24x_{11} - 64x_{11}^2 - 160x_{11}x_{13} - 64x_{11}x_{26} - 32x_{11}x_{28} \\ & - 32x_{11}x_{36} - 26x_{13} - 100x_{13}^2 - 80x_{13}x_{26} - 40x_{13}x_{28} \\ & - 40x_{13}x_{36} - 8x_{26} - 16x_{26}^2 - 16x_{26}x_{28} - 16x_{26}x_{36} - 6x_{28} \\ & - 4x_{28}^2 - 8x_{28}x_{36} - 4x_{36} - 4x_{36}^2 + x_{37} \end{aligned}$$

$$\begin{aligned} R_{22} = {} & -1 - 18x_{11} - 36x_{11}^2 - 96x_{11}x_{12} - 24x_{11}x_{22} - 24x_{12} - 64x_{12}^2 \\ & - 32x_{12}x_{22} - 6x_{22} - 4x_{22}^2 + x_{38} \end{aligned}$$

$$R_{23} = -2 + 2x_{11} - x_{11}^2 - 4x_{11}x_{38} + 4x_{38} - 4x_{38}^2 + x_{39}$$

$$R_{24} = 1 + 6x_{11} - 36x_{11}^2 - 192x_{11}x_{38} - 24x_{11}x_{39} + 16x_{38} - 256x_{38}^2 - 64x_{38}x_{39} + 2x_{39} - 4x_{39}^2 + x_{40}$$

$$R_{25} = -x_{12}^2 - 4x_{12}x_{40} - 4x_{40}^2 + x_{41}$$

$$R_{26} = -6 - 30x_{11} - 64x_{11}^2 - 128x_{11}x_{12} - 64x_{11}x_{22} - 32x_{12} - 64x_{12}^2 - 64x_{12}x_{22} - 14x_{22} - 16x_{22}^2 + x_{42}$$

$$R_{27} = -4 + 8x_{12} - 64x_{12}^2 - 96x_{12}x_{13} - 32x_{12}x_{26} - 32x_{12}x_{42} + 6x_{13} - 36x_{13}^2 - 24x_{13}x_{26} - 24x_{13}x_{42} + 2x_{26} - 4x_{26}^2 - 8x_{26}x_{42} + 4x_{42} - 4x_{42}^2 + x_{43}$$

$$R_{28} = 8x_{11} - 64x_{11}^2 - 96x_{11}x_{13} - 32x_{11}x_{28} - 32x_{11}x_{43} + 6x_{13} - 36x_{13}^2 - 24x_{13}x_{28} - 24x_{13}x_{43} + 2x_{28} - 4x_{28}^2 - 8x_{28}x_{43} + 4x_{43} - 4x_{43}^2 + x_{44}$$

$$R_{29} = -x_{13}^2 - 2x_{13}x_{14} - x_{14}^2 + x_{45}$$

$$R_{30} = 2x_{11} - x_{11}^2 - 4x_{11}x_{13} - 4x_{11}x_{45} + 4x_{13} - 4x_{13}^2 - 8x_{13}x_{45} + 4x_{45} - 4x_{45}^2 + x_{46}$$

$$R_{31} = -35 + 24x_{11} - 4x_{11}^2 - 8x_{11}x_{12} - 8x_{11}x_{26} - 8x_{11}x_{28} + 24x_{12} - 4x_{12}^2 - 8x_{12}x_{26} - 8x_{12}x_{28} + 24x_{26} - 4x_{26}^2 - 8x_{26}x_{28} + 24x_{28} - 4x_{28}^2 + x_{47}$$

$$R_{32} = -16x_{11}^2 - 32x_{11}x_{13} - 32x_{11}x_{28} - 32x_{11}x_{45} - 16x_{11}x_{46} - 16x_{11}x_{47} - 4x_{13} - 16x_{13}^2 - 32x_{13}x_{28} - 32x_{13}x_{45} - 16x_{13}x_{46} - 16x_{13}x_{47} - 16x_{28}^2 - 32x_{28}x_{45} - 16x_{28}x_{46} - 16x_{28}x_{47} - 4x_{45} - 16x_{45}^2 - 16x_{45}x_{46} - 16x_{45}x_{47} - 2x_{46} - 4x_{46}^2 - 8x_{46}x_{47} - 4x_{47}^2 + x_{48}$$

$$R_{33} = x_{13} - 25x_{13}^2 - 40x_{13}x_{28} - 20x_{13}x_{31} - 16x_{28}^2 - 16x_{28}x_{31} - 4x_{31}^2 + x_{49}$$

$$R_{34} = -1 + 8x_{12} - 4x_{12}^2 - 8x_{12}x_{26} + 8x_{26} - 4x_{26}^2 + x_{50}$$

$$R_{35} = -1 - 16x_{12}^2 - 8x_{12}x_{13} - 32x_{12}x_{26} - 16x_{12}x_{50} - x_{13}^2 - 8x_{13}x_{26} - 4x_{13}x_{50} - 16x_{26}^2 - 16x_{26}x_{50} - 4x_{50}^2 + x_{51}$$

$$R_{36} = 1 + x_{12} - x_{12}^2 - 4x_{12}x_{13} - 4x_{12}x_{51} - 4x_{13}^2 - 8x_{13}x_{51} - 4x_{51}^2 + x_{52}$$

$$R_{37} = -1 - 4x_{52} - 4x_{52}^2 + x_{53}$$

$$R_{38} = -1 - x_{13}^2 - 8x_{13}x_{52} - 4x_{13}x_{53} - 16x_{52}^2 - 16x_{52}x_{53} - 4x_{53}^2 + x_{54}$$

$$R_{39} = 2 + x_{12} - x_{12}^2 - 4x_{12}x_{13} - 4x_{12}x_{54} - 4x_{13}^2 - 8x_{13}x_{54} - 4x_{54}^2 + x_{55}$$

$$R_{40} = -4x_{49}^2 - 8x_{49}x_{55} - 4x_{55}^2 + x_{56}$$

$$R_{41} = -9 + 7x_{11} - x_{11}^2 - 8x_{11}x_{49} - 4x_{11}x_{56} + 24x_{49} - 16x_{49}^2 - 16x_{49}x_{56} + 12x_{56} - 4x_{56}^2 + x_{57}$$

$$R_{42} = -1 - 4x_{57} - 4x_{57}^2 + x_{58}$$

$$R_{43} = -x_{13}^2 - 8x_{13}x_{57} - 4x_{13}x_{58} - 16x_{57}^2 - 16x_{57}x_{58} - 4x_{58}^2 + x_{59}$$

$$R_{44} = -x_{11} - x_{11}^2 - 4x_{11}x_{13} - 4x_{11}x_{59} - 4x_{13} - 4x_{13}^2 - 8x_{13}x_{59} - 4x_{59} - 4x_{59}^2 + x_{60}$$

$$R_{45} = -10 - 12x_{60} - 4x_{60}^2 + x_{61}$$

$$R_{46} = 1 + 4x_{12} - 4x_{12}^2 - 8x_{12}x_{26} - 16x_{12}x_{60} - 8x_{12}x_{61} + 4x_{26} - 4x_{26}^2 - 16x_{26}x_{60} - 8x_{26}x_{61} + 8x_{60} - 16x_{60}^2 - 16x_{60}x_{61} + 4x_{61} - 4x_{61}^2 + x_{62}$$

$$R_{47} = x_{12} - 25x_{12}^2 - 40x_{12}x_{26} - 20x_{12}x_{62} - 16x_{26}^2 - 16x_{26}x_{62} - 4x_{62}^2 + x_{63}$$

$$R_{48} = -10 - 12x_{55} - 4x_{55}^2 + x_{64}$$

$$R_{49} = 1 - 16x_{55}^2 - 16x_{55}x_{63} - 16x_{55}x_{64} - 4x_{63}^2 - 8x_{63}x_{64} - 4x_{64}^2 + x_{65}$$

$$R_{50} = x_{13} - x_{13}^2 - 8x_{13}x_{63} - 4x_{13}x_{65} - 16x_{63}^2 - 16x_{63}x_{65} - 4x_{65}^2 + x_{66}$$

$$R_{51} = -4x_{11} - 4x_{11}^2 - 4x_{11}x_{13} - 8x_{11}x_{22} - x_{13} - x_{13}^2 - 4x_{13}x_{22} - 4x_{22} - 4x_{22}^2 + x_{67}$$

$$R_{52} = 1 + 4x_{11} - 4x_{11}^2 - 8x_{11}x_{22} - 8x_{11}x_{67} + 4x_{22} - 4x_{22}^2 - 8x_{22}x_{67} + 4x_{67} - 4x_{67}^2 + x_{68}$$

$$R_{53} = -2 - 4x_{11} - 16x_{11}^2 - 32x_{11}x_{22} - 16x_{11}x_{66} - 16x_{11}x_{68} - 4x_{22} - 16x_{22}^2 - 16x_{22}x_{66} - 16x_{22}x_{68} - 4x_{66}^2 - 8x_{66}x_{68} - 2x_{68} - 4x_{68}^2 + x_{69}$$

$$R_{54} = -2 - 4x_{69} - 4x_{69}^2 + x_{70}$$

$$R_{55} = -x_3x_{16} + x_{71}$$

$$R_{56} = x_{15} - x_{16} - 2x_{25}x_{71} - x_{72}$$

$$R_{57} = x_{15} - x_{16} - 4x_{27}x_{71} - 2x_{30}x_{71} - x_{73}$$

$$R_{58} = x_{15} - x_{16} - 2x_{32}x_{71} - 6x_{71} - x_{74}$$

$$R_{59} = x_{15} - x_{16} - 4x_{33}x_{71} - 2x_{35}x_{71} - x_{75}$$

$$R_{60} = x_{15} - x_{16} - 2x_{37}x_{71} + 6x_{71} - x_{76}$$

$$R_{61} = x_{15} - x_{16} - 4x_{40}x_{71} - 2x_{41}x_{71} - x_{77}$$

$$R_{62} = x_{15} - x_{16} - 2x_{44}x_{71} - x_{78}$$

$$R_{63} = x_{15} - x_{16} - 2x_{48}x_{71} - 4x_{71} - x_{79}$$

$$R_{64} = -x_{11}x_{12} + x_{11}x_{13} + x_{12} + x_{12}x_{13} - x_{13} - x_{13}^2 + x_{80}$$

$$R_{65} = 1 + 2x_{14} + x_{14}^2 - x_{80}^2 + x_{81}$$

$$R_{66} = -x_{81}^2 + x_{82}$$

$$R_{67} = -4x_{69}x_{71} - 2x_{70}x_{71} + 2x_{71} + x_{83}$$

$$R_{68} = -x_{15}^2 + 2x_{15}x_{16} + 2x_{15}x_{83} - x_{16}^2 - 2x_{16}x_{83} - x_{83}^2 + x_{84}$$

$$R_{69} = -1 - x_3x_{12} + x_{85}$$

$$R_{70} = -x_{16}x_{85} + x_{87}x_{88} + x_{89}$$

$$R_{71} = -1 + x_3 - x_3x_{11} + x_{86}$$

$$R_{72} = -1 + 2x_3 - 2x_3x_{11} - 2x_3x_{22} + x_{87}$$

$$R_{73} = -x_{15}x_{86} + x_{88}$$

$$R_{74} = -1 - 6x_3x_{11} - 8x_3x_{12} - 2x_3x_{22} + x_{90}$$

$$R_{75} = x_{15}x_{85} - x_{16}x_{90} + x_{91}$$

$$R_{76} = -x_{15}x_{72} + x_{16}x_{72} + x_{92}$$

$$R_{77} = -x_{73}x_{92} + x_{93}$$

$$R_{78} = -x_{74}x_{93} + x_{94}$$

$$R_{79} = -x_{75}x_{94} + x_{95}$$

$$R_{80} = -x_{76}x_{95} + x_{96}$$

$$R_{81} = -x_{77}x_{96} + x_{97}$$

$$R_{82} = -x_{78}x_{97} + x_{98}$$

$$R_{83} = -x_{79}x_{98} + x_{99}$$

$$R_{84} = -x_{89}x_{99} + x_{100}$$

$$R_{85} = -x_{91}x_{100} + x_{101}$$

$$R_{86} = -x_{101}^2 + x_{102}$$

$$R_{87} = -x_2 - x_2x_{19} + x_{19} + x_{103}$$

$$R_{88} = -x_{17}x_{103} + x_{104}$$

$$\begin{aligned} R_{89} = {} & -2x_1^2 + 2x_1x_{15} + 2x_1x_{16} + 2x_1x_{104} + 2x_1x_{105} - x_{15}^2 \\ & - 2x_{15}x_{104} - x_{16}^2 - 2x_{16}x_{105} + x_{20}x_{107} - x_{82}x_{102} - x_{84}x_{102} \\ & - x_{104}^2 - x_{105}^2 \end{aligned}$$

$$R_{90} = -x_2x_{18} + x_{18} - x_{18}x_{103} + x_{105}$$

$$\begin{aligned} R_{91} = {} & -9x_{11} - 9x_{12} - 4x_{22} - 2x_{25} - 4x_{27} - 2x_{30} - 2x_{32} - 4x_{33} - 2x_{35} - 2x_{37} \\ & - 4x_{40} - 2x_{41} - 2x_{44} - 2x_{48} - 4x_{69} - 2x_{70} + x_{106} \end{aligned}$$

$$R_{92} = -x_{108}x_{111} + x_{111} + x_{113}$$

$$R_{93} = -x_5x_{135} + x_{109}x_{235} - x_{112} - x_{113}x_{135}$$

$$R_{94} = x_{114}x_{236} - x_{116}x_{138} - x_{117}$$

$$R_{95} = x_{118}x_{236} - x_{119}x_{138} - x_{120}$$

$$R_{96} = x_{121}x_{236} - x_{122}x_{138} - x_{123}$$

$$R_{97} = x_{124}x_{236} - x_{125}x_{138} - x_{126}$$

$$R_{98} = x_{127}x_{237} - x_{129}x_{138} - x_{130}$$

$$R_{99} = x_{131}x_{237} - x_{132}x_{138} - x_{133}$$

$$R_{100} = -x_{20}x_{135} + x_{138}$$

$$R_{101} = x_{134} - x_{136}x_{138} - x_{137}$$

$$R_{102} = -x_{138}x_{140} + x_{139} - x_{141}$$

$$R_{103} = -x_{138}x_{143} + x_{142}x_{235} - x_{144}$$

$$R_{104} = -x_{138}x_{146} + x_{145}x_{236} - x_{147}$$

$$R_{105} = x_{148} - 2x_{148}x_{150} + x_{151}$$

$$R_{106} = -x_{148}x_{151} - x_{151} + x_{152}$$

$$R_{107} = 2x_{149} + x_{149}^2 + x_{151}^2 - x_{152}^2$$

$$R_{108} = -x_{149}x_{154} - x_{154} + x_{155}$$

$$R_{109} = 2x_{153} + x_{153}^2 + x_{154}^2 - x_{155}^2$$

$$R_{110} = x_4^2 - 2x_4x_{149} - 2x_{149} + x_{156}$$

$$R_{111} = -1 - x_4x_{154} + x_6 - x_{153} - x_{154} + x_{155} + x_{156}x_{157}$$

$$R_{112} = 1 - 2x_5 + x_5^2 - 2x_5x_{149} + x_{158}$$

$$R_{113} = -x_5x_{204} + x_{20} + x_{158}x_{159} - x_{203} + x_{205}$$

$$R_{114} = -1 + 2x_{149}x_{219} - x_{160} + 2x_{219} - x_{219}^2$$

$$R_{115} = -1 + x_{108} + x_{160}x_{161} - x_{203} - x_{204}x_{219} + x_{205}$$

$$R_{116} = -x_{149}x_{163} - x_{163} + x_{164}$$

$$R_{117} = 2x_{162} + x_{162}^2 + x_{163}^2 - x_{164}^2$$

$$R_{118} = 1 - 2x_{108} + x_{108}^2 - 2x_{108}x_{149} + x_{165}$$

$$R_{119} = -1 - x_{108}x_{163} + x_{109} - x_{162} + x_{164} + x_{165}x_{166}$$

$$R_{120} = 4 - 4x_{108} + x_{108}^2 - 2x_{108}x_{149} + 2x_{149} + x_{167}$$

$$R_{121} = -1 - x_{108}x_{154} + x_{110} - x_{153} + x_{154} + x_{155} + x_{167}x_{168}$$

$$R_{122} = 1 + x_{149} - x_{149}x_{170} - x_{170} + x_{171}$$

$$R_{123} = 1 + 2x_{169} + x_{169}^2 - 2x_{170} + x_{170}^2 - x_{171}^2$$

$$R_{124} = -1 + x_{108} - x_{108}x_{170} + x_{114} + x_{165}x_{172} - x_{169} + x_{171}$$

$$R_{125} = -2x_{149} - x_{149}^2 + x_{173}$$

$$\begin{aligned} R_{126} = 1 &- x_{108} - x_{108}x_{153} - x_{108}x_{155} + x_{115} + x_{149}x_{155} + x_{153} - x_{154}x_{173} \\ &+ 2x_{155} + x_{167}x_{174} \end{aligned}$$

$$R_{127} = -x_{149}x_{176} - x_{176} + x_{177}$$

$$R_{128} = 2x_{175} + x_{175}^2 + x_{176}^2 - x_{177}^2$$

$$R_{129} = -1 - x_{108}x_{176} + x_{118} + x_{165}x_{178} - x_{175} + x_{177}$$

$$R_{130} = -x_{149}x_{180} - x_{180} + x_{181}$$

$$R_{131} = 2x_{179} + x_{179}^2 + x_{180}^2 - x_{181}^2$$

$$R_{132} = -1 - x_{108}x_{180} + x_{121} + x_{165}x_{182} - x_{179} + x_{181}$$

$$R_{133} = -x_{149}x_{184} - x_{184} + x_{185}$$

$$R_{134} = 2x_{183} + x_{183}^2 + x_{184}^2 - x_{185}^2$$

$$R_{135} = -1 - x_{108}x_{184} + x_{124} + x_{165}x_{186} - x_{183} + x_{185}$$

$$R_{136} = -x_{149}x_{188} - x_{188} + x_{189}$$

$$R_{137} = 2x_{187} + x_{187}^2 + x_{188}^2 - x_{189}^2$$

$$R_{138} = -1 - x_{108}x_{188} + x_{127} + x_{165}x_{190} - x_{187} + x_{189}$$

$$R_{139} = -x_{149}x_{192} - x_{192} + x_{193}$$

$$R_{140} = 2x_{191} + x_{191}^2 + x_{192}^2 - x_{193}^2$$

$$R_{141} = -1 - x_{108}x_{192} + x_{128} + x_{167}x_{194} - x_{191} + x_{192} + x_{193}$$

$$R_{142} = -x_{149}x_{196} - x_{196} + x_{197}$$

$$R_{143} = 2x_{195} + x_{195}^2 + x_{196}^2 - x_{197}^2$$

$$R_{144} = -1 - x_{108}x_{196} + x_{131} + x_{165}x_{198} - x_{195} + x_{197}$$

$$R_{145} = -x_{149}x_{200} - x_{200} + x_{201}$$

$$R_{146} = 2x_{199} + x_{199}^2 + x_{200}^2 - x_{201}^2$$

$$R_{147} = -1 - x_{108}x_{200} + x_{134} + x_{165}x_{202} - x_{199} + x_{201}$$

$$R_{148} = 1 - x_{149}x_{204} - x_{204} + x_{205}$$

$$R_{149} = -1 + 2x_{203} + x_{203}^2 + x_{204}^2 - 2x_{205} - x_{205}^2$$

$$R_{150} = -x_{108}x_{204} + x_{135} + x_{167}x_{206} - x_{203} + x_{204} + x_{205}$$

$$R_{151} = -x_{149}x_{208} - x_{208} + x_{209}$$

$$R_{152} = 2x_{207} + x_{207}^2 + x_{208}^2 - x_{209}^2$$

$$R_{153} = -1 - x_{108}x_{208} + x_{139} + x_{165}x_{210} - x_{207} + x_{209}$$

$$R_{154} = -x_{149}x_{212} - x_{212} + x_{213}$$

$$R_{155} = 2x_{211} + x_{211}^2 + x_{212}^2 - x_{213}^2$$

$$R_{156} = -1 - x_{108}x_{212} + x_{142} + x_{165}x_{214} - x_{211} + x_{213}$$

$$R_{157} = -x_{149}x_{216} - x_{216} + x_{217}$$

$$R_{158} = 2x_{215} + x_{215}^2 + x_{216}^2 - x_{217}^2$$

$$R_{159} = -1 - x_{108}x_{216} + x_{145} + x_{165}x_{218} - x_{215} + x_{217}$$

$$\begin{aligned} R_{160} = {} & -6 - 2x_0 - 3x_0^2 - 2x_0x_4 - x_1^2 - x_2^2 - 2x_4 - 2x_4^2 - x_5^2 \\ & - x_6^2 + 2x_{11} - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 \\ & - x_{17}^2 - x_{18}^2 - x_{19}^2 - x_{20}^2 - x_{106}^2 + 2x_{108} - 2x_{108}^2 - x_{109}^2 \\ & - x_{110}^2 - x_{114}^2 - x_{115}^2 - x_{118}^2 - x_{121}^2 - x_{124}^2 - x_{127}^2 \\ & - x_{128}^2 - x_{131}^2 - x_{134}^2 - x_{135}^2 - x_{139}^2 - x_{142}^2 - x_{145}^2 + x_{148} \\ & - x_{219}^2 \end{aligned}$$

$$\begin{aligned} R_{161} = {} & -x_5x_{11} - x_5x_{12} - x_5x_{13} - x_5x_{14} - x_5x_{15} - x_5x_{16} - x_5x_{17} \\ & - x_5x_{18} - x_5x_{19} + x_{219} \end{aligned}$$

$$R_{162} = -16 + x_0 - x_{220}^2$$

$$R_{163} = -1 + x_2 + x_6 - x_6^2$$

$$R_{164} = -x_{149}x_{222} - x_{222} + x_{231}$$

$$R_{165} = -1 + x_{221}^2 + x_{222}^2 - x_{231}^2$$

$$R_{166} = -x_{224} - x_{224}x_{225} + x_{232}$$

$$R_{167} = -1 + x_{223}^2 + x_{224}^2 - x_{232}^2$$

$$R_{168} = -x_{216}^2 + x_{233}$$

$$R_{169} = x_{222} - 2x_{227}x_{233} - x_{233}$$

$$R_{170} = -4x_{222} - 8x_{222}x_{226} + x_{225}$$

$$R_{171} = -x_{149} - x_{221}x_{228} + x_{225}$$

$$R_{172} = -1 - x_{215} - x_{221}x_{229} + x_{223}$$

$$R_{173} = -x_{106} - 4x_{216}x_{230} + x_{224}$$

$$\begin{aligned} R_{174} = {} & 2x_0^2 + 2x_0x_4 - 2x_0x_{154} - 2x_0x_{163} + x_1^2 - 2x_1x_{204} \\ & + x_2^2 - 2x_2x_{192} + x_4^2 - 2x_4x_{163} + x_{11}^2 - 2x_{11}x_{170} + x_{12}^2 \\ & - 2x_{12}x_{176} + x_{13}^2 - 2x_{13}x_{180} + x_{14}^2 - 2x_{14}x_{184} + x_{15}^2 \\ & - 2x_{15}x_{188} + x_{16}^2 - 2x_{16}x_{196} + x_{17}^2 - 2x_{17}x_{200} + x_{18}^2 \\ & - 2x_{18}x_{208} + x_{19}^2 - 2x_{19}x_{212} + x_{106}^2 - 2x_{106}x_{216} - x_{149}x_{234} \\ & + x_{153}^2 + x_{154}^2 + x_{162}^2 + x_{163}^2 + x_{169}^2 + x_{170}^2 + x_{175}^2 \\ & + x_{176}^2 + x_{179}^2 + x_{180}^2 + x_{183}^2 + x_{184}^2 + x_{187}^2 + x_{188}^2 \\ & + x_{191}^2 + x_{192}^2 + x_{195}^2 + x_{196}^2 + x_{199}^2 + x_{200}^2 + x_{203}^2 \\ & + x_{204}^2 + x_{207}^2 + x_{208}^2 + x_{211}^2 + x_{212}^2 + x_{215}^2 + x_{216}^2 \end{aligned}$$

$$R_{175} = -x_{110}x_{235} + x_{135}$$

$$R_{176} = -x_{115}x_{236} + x_{135}$$

$$R_{177} = -x_{128}x_{237} + x_{135}$$

$$R_{178} = -2 + x_{108} - 4x_{238} - 4x_{238}^2$$

$$R_{179} = -1 + x_5 + x_5x_7 - x_7 - x_7^2 - x_{239}^2 - x_{240}^2 - x_{241} - x_{241}^2$$

$$\begin{aligned} R_{180} = {} & -2 - x_{106}^2 - x_{154}^2 - x_{163}^2 + 2x_{170} - x_{170}^2 - x_{176}^2 - x_{180}^2 \\ & - x_{184}^2 - x_{188}^2 - x_{192}^2 - x_{196}^2 - x_{200}^2 - x_{204}^2 - x_{208}^2 \\ & - x_{212}^2 + x_{216}^2 - x_{242}^2 - x_{243}^2 - x_{244} - x_{244}^2 \end{aligned}$$

$$\begin{aligned} R_{181} = {} & -1 - 4x_{112}^2 - 4x_{117}^2 - 4x_{120}^2 - 4x_{123}^2 - 4x_{126}^2 - 4x_{130}^2 \\ & - 4x_{133}^2 + x_{135}^2 - 4x_{137}^2 - 4x_{141}^2 - 4x_{144}^2 - 4x_{147}^2 \\ & - x_{245}^2 - x_{246}^2 - x_{247} - x_{247}^2 \end{aligned}$$

C Complete monic-elimination pivot tables

Every entry $j \mapsto v$ selects the source residual R_j and source coordinate x_v of the printed 248-variable equation. Both indices are zero based: $0 \le j < 182$ and $0 \le v < 248$. Write the selected row uniquely as $R_j = \varepsilon_v x_v + f_v$ with $\varepsilon_v = \pm 1$, and replace x_v by $-\varepsilon_v f_v$. The substitution dependency graph is acyclic; apply the definitions recursively, evaluating dependencies before their users (a reverse topological order for the graph defined in the theorem). Retain, in increasing source-index order, every coordinate not selected as a target. The new polynomial is the sum of the squares of all unselected residuals after these substitutions. Thus these finite tables and the complete source residual list specify all eight polynomials without a search rule or unspecified compiler choice. The machine-readable certificates and literal polynomial DAGs supply the same definitions independently.

C.1 225 integer variables, degree 6

There are 23 selected coordinates and 159 remaining residuals.

Source residual $j \mapsto$ eliminated coordinate v					
$2 \mapsto 10$	$56 \mapsto 72$	$69 \mapsto 85$	$73 \mapsto 88$	$74 \mapsto 90$	$77 \mapsto 93$
$79 \mapsto 95$	$81 \mapsto 97$	$83 \mapsto 99$	$86 \mapsto 102$	$87 \mapsto 103$	$91 \mapsto 106$
$92 \mapsto 113$	$100 \mapsto 138$	$110 \mapsto 156$	$112 \mapsto 158$	$114 \mapsto 160$	$118 \mapsto 165$
$120 \mapsto 167$	$125 \mapsto 173$	$160 \mapsto 148$	$168 \mapsto 233$	$170 \mapsto 225$	

C.2 153 integer variables, degree 8

There are 95 selected coordinates and 87 remaining residuals.

Source residual $j \mapsto$ eliminated coordinate v					
$1 \mapsto 6$	$2 \mapsto 10$	$3 \mapsto 1$	$5 \mapsto 19$	$8 \mapsto 24$	$11 \mapsto 27$
$13 \mapsto 29$	$15 \mapsto 31$	$17 \mapsto 33$	$18 \mapsto 34$	$20 \mapsto 36$	$23 \mapsto 39$
$25 \mapsto 41$	$26 \mapsto 42$	$30 \mapsto 46$	$31 \mapsto 47$	$34 \mapsto 50$	$37 \mapsto 53$
$40 \mapsto 56$	$42 \mapsto 58$	$45 \mapsto 61$	$47 \mapsto 63$	$48 \mapsto 64$	$51 \mapsto 67$
$54 \mapsto 70$	$56 \mapsto 72$	$58 \mapsto 74$	$60 \mapsto 76$	$62 \mapsto 78$	$64 \mapsto 80$
$66 \mapsto 82$	$68 \mapsto 84$	$69 \mapsto 85$	$72 \mapsto 87$	$73 \mapsto 88$	$74 \mapsto 90$
$77 \mapsto 93$	$79 \mapsto 95$	$81 \mapsto 97$	$83 \mapsto 99$	$84 \mapsto 100$	$86 \mapsto 102$
$88 \mapsto 104$	$90 \mapsto 105$	$91 \mapsto 106$	$92 \mapsto 113$	$94 \mapsto 117$	$95 \mapsto 120$
$96 \mapsto 123$	$97 \mapsto 126$	$98 \mapsto 130$	$99 \mapsto 133$	$101 \mapsto 137$	$102 \mapsto 141$
$103 \mapsto 144$	$104 \mapsto 147$	$106 \mapsto 152$	$108 \mapsto 155$	$110 \mapsto 156$	$112 \mapsto 158$
$114 \mapsto 160$	$116 \mapsto 164$	$119 \mapsto 162$	$121 \mapsto 153$	$122 \mapsto 171$	$124 \mapsto 169$
$125 \mapsto 173$	$127 \mapsto 177$	$129 \mapsto 175$	$130 \mapsto 181$	$132 \mapsto 179$	$133 \mapsto 185$
$135 \mapsto 183$	$136 \mapsto 189$	$138 \mapsto 187$	$139 \mapsto 193$	$141 \mapsto 191$	$142 \mapsto 197$
$144 \mapsto 195$	$145 \mapsto 201$	$147 \mapsto 199$	$148 \mapsto 205$	$150 \mapsto 203$	$151 \mapsto 209$
$153 \mapsto 207$	$154 \mapsto 213$	$156 \mapsto 211$	$157 \mapsto 217$	$162 \mapsto 0$	$164 \mapsto 231$
$168 \mapsto 233$	$170 \mapsto 225$	$172 \mapsto 215$	$173 \mapsto 224$	$175 \mapsto 135$	

C.3 144 integer variables, degree 12

There are 104 selected coordinates and 78 remaining residuals.

Source residual $j \mapsto$ eliminated coordinate v					
$0 \mapsto 4$	$1 \mapsto 6$	$2 \mapsto 10$	$3 \mapsto 1$	$5 \mapsto 19$	$8 \mapsto 24$
$13 \mapsto 29$	$16 \mapsto 32$	$17 \mapsto 33$	$18 \mapsto 34$	$20 \mapsto 36$	$23 \mapsto 39$
$26 \mapsto 42$	$28 \mapsto 44$	$30 \mapsto 46$	$31 \mapsto 47$	$34 \mapsto 50$	$37 \mapsto 53$
$40 \mapsto 56$	$42 \mapsto 58$	$45 \mapsto 61$	$47 \mapsto 63$	$48 \mapsto 64$	$51 \mapsto 67$
$56 \mapsto 72$	$57 \mapsto 73$	$58 \mapsto 74$	$59 \mapsto 75$	$60 \mapsto 76$	$61 \mapsto 77$
$62 \mapsto 78$	$63 \mapsto 79$	$64 \mapsto 80$	$66 \mapsto 82$	$67 \mapsto 83$	$68 \mapsto 84$

Source residual $j \mapsto$ eliminated coordinate v (continued)					
$69 \mapsto 85$	$71 \mapsto 86$	$72 \mapsto 87$	$73 \mapsto 88$	$74 \mapsto 90$	$77 \mapsto 93$
$79 \mapsto 95$	$81 \mapsto 97$	$83 \mapsto 99$	$84 \mapsto 100$	$86 \mapsto 102$	$88 \mapsto 104$
$90 \mapsto 105$	$91 \mapsto 106$	$92 \mapsto 113$	$93 \mapsto 112$	$94 \mapsto 117$	$95 \mapsto 120$
$96 \mapsto 123$	$97 \mapsto 126$	$98 \mapsto 130$	$99 \mapsto 133$	$100 \mapsto 138$	$101 \mapsto 137$
$102 \mapsto 141$	$103 \mapsto 144$	$104 \mapsto 147$	$105 \mapsto 151$	$106 \mapsto 152$	$108 \mapsto 155$
$110 \mapsto 156$	$112 \mapsto 158$	$114 \mapsto 160$	$116 \mapsto 164$	$119 \mapsto 162$	$121 \mapsto 153$
$122 \mapsto 171$	$124 \mapsto 114$	$125 \mapsto 173$	$126 \mapsto 115$	$127 \mapsto 177$	$129 \mapsto 175$
$130 \mapsto 181$	$132 \mapsto 179$	$133 \mapsto 185$	$135 \mapsto 183$	$136 \mapsto 189$	$138 \mapsto 187$
$139 \mapsto 193$	$141 \mapsto 191$	$142 \mapsto 197$	$144 \mapsto 195$	$145 \mapsto 201$	$147 \mapsto 199$
$148 \mapsto 205$	$150 \mapsto 203$	$151 \mapsto 209$	$153 \mapsto 207$	$154 \mapsto 213$	$156 \mapsto 211$
$157 \mapsto 217$	$159 \mapsto 145$	$162 \mapsto 0$	$164 \mapsto 231$	$168 \mapsto 233$	$170 \mapsto 225$
$172 \mapsto 215$	$173 \mapsto 224$				

C.4 125 integer variables, degree 16

There are 123 selected coordinates and 59 remaining residuals.

Source residual $j \mapsto$ eliminated coordinate v					
$0 \mapsto 4$	$1 \mapsto 6$	$2 \mapsto 10$	$3 \mapsto 1$	$5 \mapsto 19$	$7 \mapsto 23$
$8 \mapsto 24$	$13 \mapsto 29$	$16 \mapsto 32$	$17 \mapsto 33$	$18 \mapsto 34$	$20 \mapsto 36$
$22 \mapsto 38$	$23 \mapsto 39$	$25 \mapsto 41$	$26 \mapsto 42$	$28 \mapsto 44$	$29 \mapsto 45$
$30 \mapsto 46$	$31 \mapsto 47$	$33 \mapsto 49$	$34 \mapsto 50$	$35 \mapsto 51$	$37 \mapsto 53$
$38 \mapsto 54$	$40 \mapsto 56$	$42 \mapsto 58$	$43 \mapsto 59$	$45 \mapsto 61$	$47 \mapsto 63$
$48 \mapsto 64$	$49 \mapsto 65$	$51 \mapsto 67$	$52 \mapsto 68$	$54 \mapsto 70$	$56 \mapsto 72$
$57 \mapsto 73$	$58 \mapsto 74$	$59 \mapsto 75$	$60 \mapsto 76$	$61 \mapsto 77$	$62 \mapsto 78$
$63 \mapsto 79$	$64 \mapsto 80$	$66 \mapsto 82$	$67 \mapsto 83$	$68 \mapsto 84$	$69 \mapsto 85$
$71 \mapsto 86$	$72 \mapsto 87$	$73 \mapsto 88$	$74 \mapsto 90$	$75 \mapsto 91$	$76 \mapsto 92$
$77 \mapsto 93$	$79 \mapsto 95$	$81 \mapsto 97$	$83 \mapsto 99$	$84 \mapsto 100$	$86 \mapsto 102$
$87 \mapsto 103$	$88 \mapsto 104$	$90 \mapsto 105$	$91 \mapsto 106$	$92 \mapsto 113$	$93 \mapsto 112$
$94 \mapsto 117$	$95 \mapsto 120$	$96 \mapsto 123$	$97 \mapsto 126$	$98 \mapsto 130$	$99 \mapsto 133$
$100 \mapsto 138$	$101 \mapsto 137$	$102 \mapsto 141$	$103 \mapsto 144$	$104 \mapsto 147$	$105 \mapsto 151$
$106 \mapsto 152$	$108 \mapsto 155$	$110 \mapsto 156$	$112 \mapsto 158$	$114 \mapsto 160$	$116 \mapsto 164$
$118 \mapsto 165$	$119 \mapsto 162$	$120 \mapsto 167$	$121 \mapsto 153$	$122 \mapsto 171$	$124 \mapsto 114$
$125 \mapsto 173$	$126 \mapsto 115$	$127 \mapsto 177$	$129 \mapsto 175$	$130 \mapsto 181$	$132 \mapsto 179$
$133 \mapsto 185$	$135 \mapsto 183$	$136 \mapsto 189$	$138 \mapsto 187$	$139 \mapsto 193$	$141 \mapsto 191$
$142 \mapsto 197$	$144 \mapsto 195$	$145 \mapsto 201$	$147 \mapsto 199$	$148 \mapsto 205$	$150 \mapsto 203$
$151 \mapsto 209$	$153 \mapsto 207$	$154 \mapsto 213$	$156 \mapsto 211$	$157 \mapsto 217$	$159 \mapsto 145$
$161 \mapsto 219$	$162 \mapsto 0$	$164 \mapsto 231$	$166 \mapsto 232$	$168 \mapsto 233$	$170 \mapsto 225$
$172 \mapsto 215$	$173 \mapsto 224$	$175 \mapsto 135$			

C.5 117 integer variables, degree 32

There are 131 selected coordinates and 51 remaining residuals.

Source residual $j \mapsto$ eliminated coordinate v					
$0 \mapsto 4$	$1 \mapsto 6$	$2 \mapsto 10$	$3 \mapsto 1$	$5 \mapsto 19$	$6 \mapsto 22$
$7 \mapsto 23$	$8 \mapsto 24$	$13 \mapsto 29$	$15 \mapsto 31$	$18 \mapsto 34$	$20 \mapsto 36$
$22 \mapsto 38$	$23 \mapsto 39$	$26 \mapsto 42$	$28 \mapsto 44$	$29 \mapsto 45$	$30 \mapsto 46$
$31 \mapsto 47$	$33 \mapsto 49$	$34 \mapsto 50$	$35 \mapsto 51$	$37 \mapsto 53$	$38 \mapsto 54$
$40 \mapsto 56$	$42 \mapsto 58$	$43 \mapsto 59$	$45 \mapsto 61$	$47 \mapsto 63$	$48 \mapsto 64$
$49 \mapsto 65$	$50 \mapsto 66$	$51 \mapsto 67$	$52 \mapsto 68$	$54 \mapsto 70$	$56 \mapsto 72$
$57 \mapsto 73$	$58 \mapsto 74$	$59 \mapsto 75$	$60 \mapsto 76$	$61 \mapsto 77$	$62 \mapsto 78$
$63 \mapsto 79$	$64 \mapsto 80$	$65 \mapsto 81$	$66 \mapsto 82$	$67 \mapsto 83$	$68 \mapsto 84$
$69 \mapsto 85$	$70 \mapsto 89$	$71 \mapsto 86$	$72 \mapsto 87$	$73 \mapsto 88$	$74 \mapsto 90$
$75 \mapsto 91$	$76 \mapsto 92$	$77 \mapsto 93$	$78 \mapsto 94$	$79 \mapsto 95$	$80 \mapsto 96$

Source residual $j \mapsto$ eliminated coordinate v (continued)					
$81 \mapsto 97$	$83 \mapsto 99$	$84 \mapsto 100$	$86 \mapsto 102$	$87 \mapsto 103$	$88 \mapsto 104$
$90 \mapsto 105$	$91 \mapsto 106$	$92 \mapsto 113$	$93 \mapsto 112$	$94 \mapsto 117$	$95 \mapsto 120$
$96 \mapsto 123$	$97 \mapsto 126$	$98 \mapsto 130$	$99 \mapsto 133$	$100 \mapsto 138$	$101 \mapsto 137$
$102 \mapsto 141$	$103 \mapsto 144$	$104 \mapsto 147$	$105 \mapsto 151$	$106 \mapsto 152$	$108 \mapsto 155$
$110 \mapsto 156$	$112 \mapsto 158$	$113 \mapsto 20$	$114 \mapsto 160$	$116 \mapsto 164$	$118 \mapsto 165$
$119 \mapsto 162$	$120 \mapsto 167$	$121 \mapsto 153$	$122 \mapsto 171$	$124 \mapsto 114$	$125 \mapsto 173$
$126 \mapsto 115$	$127 \mapsto 177$	$129 \mapsto 175$	$130 \mapsto 181$	$132 \mapsto 179$	$133 \mapsto 185$
$135 \mapsto 183$	$136 \mapsto 189$	$138 \mapsto 187$	$139 \mapsto 193$	$141 \mapsto 191$	$142 \mapsto 197$
$144 \mapsto 195$	$145 \mapsto 201$	$147 \mapsto 199$	$148 \mapsto 205$	$150 \mapsto 203$	$151 \mapsto 209$
$153 \mapsto 207$	$154 \mapsto 213$	$156 \mapsto 211$	$157 \mapsto 217$	$159 \mapsto 145$	$161 \mapsto 219$
$162 \mapsto 0$	$163 \mapsto 2$	$164 \mapsto 231$	$166 \mapsto 232$	$168 \mapsto 233$	$169 \mapsto 222$
$170 \mapsto 225$	$172 \mapsto 215$	$173 \mapsto 224$	$175 \mapsto 135$	$178 \mapsto 108$	

C.6 111 integer variables, degree 64

There are 137 selected coordinates and 45 remaining residuals.

Source residual $j \mapsto$ eliminated coordinate v					
$0 \mapsto 4$	$1 \mapsto 6$	$2 \mapsto 10$	$3 \mapsto 1$	$5 \mapsto 19$	$7 \mapsto 23$
$8 \mapsto 24$	$11 \mapsto 27$	$13 \mapsto 29$	$16 \mapsto 32$	$17 \mapsto 33$	$18 \mapsto 34$
$19 \mapsto 35$	$20 \mapsto 36$	$22 \mapsto 38$	$23 \mapsto 39$	$25 \mapsto 41$	$26 \mapsto 42$
$28 \mapsto 44$	$29 \mapsto 45$	$30 \mapsto 46$	$31 \mapsto 47$	$33 \mapsto 49$	$34 \mapsto 50$
$35 \mapsto 51$	$37 \mapsto 53$	$38 \mapsto 54$	$40 \mapsto 56$	$42 \mapsto 58$	$43 \mapsto 59$
$44 \mapsto 60$	$45 \mapsto 61$	$47 \mapsto 63$	$48 \mapsto 64$	$49 \mapsto 65$	$50 \mapsto 66$
$51 \mapsto 67$	$52 \mapsto 68$	$54 \mapsto 70$	$56 \mapsto 72$	$57 \mapsto 73$	$58 \mapsto 74$
$59 \mapsto 75$	$60 \mapsto 76$	$61 \mapsto 77$	$62 \mapsto 78$	$63 \mapsto 79$	$64 \mapsto 80$
$65 \mapsto 81$	$66 \mapsto 82$	$67 \mapsto 83$	$68 \mapsto 84$	$69 \mapsto 85$	$70 \mapsto 89$
$71 \mapsto 86$	$72 \mapsto 87$	$73 \mapsto 88$	$74 \mapsto 90$	$75 \mapsto 91$	$76 \mapsto 92$
$77 \mapsto 93$	$78 \mapsto 94$	$79 \mapsto 95$	$80 \mapsto 96$	$81 \mapsto 97$	$82 \mapsto 98$
$83 \mapsto 99$	$84 \mapsto 100$	$86 \mapsto 102$	$87 \mapsto 103$	$88 \mapsto 104$	$90 \mapsto 105$
$91 \mapsto 106$	$92 \mapsto 113$	$93 \mapsto 112$	$94 \mapsto 117$	$95 \mapsto 120$	$96 \mapsto 123$
$97 \mapsto 126$	$98 \mapsto 130$	$99 \mapsto 133$	$100 \mapsto 138$	$101 \mapsto 137$	$102 \mapsto 141$
$103 \mapsto 144$	$104 \mapsto 147$	$105 \mapsto 151$	$106 \mapsto 152$	$108 \mapsto 155$	$110 \mapsto 156$
$112 \mapsto 158$	$113 \mapsto 20$	$114 \mapsto 160$	$116 \mapsto 164$	$118 \mapsto 165$	$119 \mapsto 162$
$120 \mapsto 167$	$121 \mapsto 153$	$122 \mapsto 171$	$124 \mapsto 114$	$125 \mapsto 173$	$126 \mapsto 115$
$127 \mapsto 177$	$129 \mapsto 175$	$130 \mapsto 181$	$132 \mapsto 179$	$133 \mapsto 185$	$135 \mapsto 183$
$136 \mapsto 189$	$138 \mapsto 187$	$139 \mapsto 193$	$141 \mapsto 191$	$142 \mapsto 197$	$144 \mapsto 195$
$145 \mapsto 201$	$147 \mapsto 199$	$148 \mapsto 205$	$150 \mapsto 203$	$151 \mapsto 209$	$153 \mapsto 207$
$154 \mapsto 213$	$156 \mapsto 211$	$157 \mapsto 217$	$159 \mapsto 145$	$161 \mapsto 219$	$162 \mapsto 0$
$163 \mapsto 2$	$164 \mapsto 231$	$166 \mapsto 232$	$168 \mapsto 233$	$169 \mapsto 222$	$170 \mapsto 225$
$171 \mapsto 149$	$172 \mapsto 215$	$173 \mapsto 224$	$175 \mapsto 135$	$178 \mapsto 108$	

C.7 102 integer variables, degree 256

There are 146 selected coordinates and 36 remaining residuals.

Source residual $j \mapsto$ eliminated coordinate v					
$0 \mapsto 4$	$1 \mapsto 6$	$2 \mapsto 10$	$3 \mapsto 1$	$5 \mapsto 19$	$7 \mapsto 23$
$8 \mapsto 24$	$11 \mapsto 27$	$13 \mapsto 29$	$14 \mapsto 30$	$15 \mapsto 31$	$16 \mapsto 32$
$17 \mapsto 33$	$18 \mapsto 34$	$19 \mapsto 35$	$20 \mapsto 36$	$21 \mapsto 37$	$22 \mapsto 38$
$23 \mapsto 39$	$25 \mapsto 41$	$26 \mapsto 42$	$27 \mapsto 43$	$28 \mapsto 44$	$29 \mapsto 45$
$30 \mapsto 46$	$31 \mapsto 47$	$32 \mapsto 48$	$33 \mapsto 49$	$34 \mapsto 50$	$35 \mapsto 51$
$36 \mapsto 52$	$37 \mapsto 53$	$38 \mapsto 54$	$40 \mapsto 56$	$42 \mapsto 58$	$43 \mapsto 59$
$44 \mapsto 60$	$45 \mapsto 61$	$47 \mapsto 63$	$48 \mapsto 64$	$49 \mapsto 65$	$50 \mapsto 66$
$51 \mapsto 67$	$52 \mapsto 68$	$54 \mapsto 70$	$55 \mapsto 71$	$56 \mapsto 72$	$57 \mapsto 73$

Source residual $j \mapsto$ eliminated coordinate v (continued)					
$58 \mapsto 74$	$59 \mapsto 75$	$60 \mapsto 76$	$61 \mapsto 77$	$62 \mapsto 78$	$63 \mapsto 79$
$64 \mapsto 80$	$65 \mapsto 81$	$66 \mapsto 82$	$67 \mapsto 83$	$68 \mapsto 84$	$69 \mapsto 85$
$70 \mapsto 89$	$71 \mapsto 86$	$72 \mapsto 87$	$73 \mapsto 88$	$74 \mapsto 90$	$75 \mapsto 91$
$76 \mapsto 92$	$77 \mapsto 93$	$78 \mapsto 94$	$79 \mapsto 95$	$80 \mapsto 96$	$81 \mapsto 97$
$82 \mapsto 98$	$83 \mapsto 99$	$84 \mapsto 100$	$85 \mapsto 101$	$86 \mapsto 102$	$87 \mapsto 103$
$88 \mapsto 104$	$90 \mapsto 105$	$91 \mapsto 106$	$92 \mapsto 113$	$93 \mapsto 112$	$94 \mapsto 117$
$95 \mapsto 120$	$96 \mapsto 123$	$97 \mapsto 126$	$98 \mapsto 130$	$99 \mapsto 133$	$100 \mapsto 138$
$101 \mapsto 137$	$102 \mapsto 141$	$103 \mapsto 144$	$104 \mapsto 147$	$105 \mapsto 151$	$106 \mapsto 152$
$108 \mapsto 155$	$110 \mapsto 156$	$112 \mapsto 158$	$113 \mapsto 20$	$114 \mapsto 160$	$116 \mapsto 164$
$118 \mapsto 165$	$119 \mapsto 162$	$120 \mapsto 167$	$121 \mapsto 153$	$122 \mapsto 171$	$124 \mapsto 114$
$125 \mapsto 173$	$126 \mapsto 115$	$127 \mapsto 177$	$129 \mapsto 175$	$130 \mapsto 181$	$132 \mapsto 179$
$133 \mapsto 185$	$135 \mapsto 183$	$136 \mapsto 189$	$138 \mapsto 187$	$139 \mapsto 193$	$141 \mapsto 191$
$142 \mapsto 197$	$144 \mapsto 195$	$145 \mapsto 201$	$147 \mapsto 199$	$148 \mapsto 205$	$150 \mapsto 203$
$151 \mapsto 209$	$153 \mapsto 207$	$154 \mapsto 213$	$156 \mapsto 211$	$157 \mapsto 217$	$159 \mapsto 145$
$160 \mapsto 148$	$161 \mapsto 219$	$162 \mapsto 0$	$163 \mapsto 2$	$164 \mapsto 231$	$166 \mapsto 232$
$168 \mapsto 233$	$169 \mapsto 222$	$170 \mapsto 225$	$171 \mapsto 149$	$172 \mapsto 215$	$173 \mapsto 224$
$175 \mapsto 135$	$178 \mapsto 108$				

C.8 94 integer variables, degree 65,540

There are 154 selected coordinates and 28 remaining residuals.

Source residual $j \mapsto$ eliminated coordinate v					
$0 \mapsto 4$	$1 \mapsto 6$	$2 \mapsto 10$	$3 \mapsto 1$	$5 \mapsto 19$	$6 \mapsto 22$
$7 \mapsto 23$	$8 \mapsto 24$	$9 \mapsto 25$	$10 \mapsto 26$	$11 \mapsto 27$	$12 \mapsto 28$
$13 \mapsto 29$	$14 \mapsto 30$	$15 \mapsto 31$	$16 \mapsto 32$	$17 \mapsto 33$	$18 \mapsto 34$
$19 \mapsto 35$	$20 \mapsto 36$	$21 \mapsto 37$	$22 \mapsto 38$	$23 \mapsto 39$	$24 \mapsto 40$
$25 \mapsto 41$	$26 \mapsto 42$	$27 \mapsto 43$	$28 \mapsto 44$	$29 \mapsto 45$	$30 \mapsto 46$
$31 \mapsto 47$	$32 \mapsto 48$	$33 \mapsto 49$	$34 \mapsto 50$	$35 \mapsto 51$	$36 \mapsto 52$
$37 \mapsto 53$	$38 \mapsto 54$	$39 \mapsto 55$	$40 \mapsto 56$	$42 \mapsto 58$	$43 \mapsto 59$
$44 \mapsto 60$	$45 \mapsto 61$	$46 \mapsto 62$	$47 \mapsto 63$	$48 \mapsto 64$	$49 \mapsto 65$
$50 \mapsto 66$	$51 \mapsto 67$	$52 \mapsto 68$	$53 \mapsto 69$	$54 \mapsto 70$	$55 \mapsto 71$
$56 \mapsto 72$	$57 \mapsto 73$	$58 \mapsto 74$	$59 \mapsto 75$	$60 \mapsto 76$	$61 \mapsto 77$
$62 \mapsto 78$	$63 \mapsto 79$	$64 \mapsto 80$	$65 \mapsto 81$	$66 \mapsto 82$	$67 \mapsto 83$
$68 \mapsto 84$	$69 \mapsto 85$	$70 \mapsto 89$	$71 \mapsto 86$	$72 \mapsto 87$	$73 \mapsto 88$
$74 \mapsto 90$	$75 \mapsto 91$	$76 \mapsto 92$	$77 \mapsto 93$	$78 \mapsto 94$	$79 \mapsto 95$
$80 \mapsto 96$	$81 \mapsto 97$	$82 \mapsto 98$	$83 \mapsto 99$	$84 \mapsto 100$	$85 \mapsto 101$
$86 \mapsto 102$	$87 \mapsto 103$	$88 \mapsto 104$	$90 \mapsto 105$	$91 \mapsto 106$	$92 \mapsto 113$
$93 \mapsto 112$	$94 \mapsto 117$	$95 \mapsto 120$	$96 \mapsto 123$	$97 \mapsto 126$	$98 \mapsto 130$
$99 \mapsto 133$	$100 \mapsto 138$	$101 \mapsto 137$	$102 \mapsto 141$	$103 \mapsto 144$	$104 \mapsto 147$
$105 \mapsto 151$	$106 \mapsto 152$	$108 \mapsto 155$	$110 \mapsto 156$	$112 \mapsto 158$	$113 \mapsto 20$
$114 \mapsto 160$	$116 \mapsto 164$	$118 \mapsto 165$	$119 \mapsto 162$	$120 \mapsto 167$	$121 \mapsto 153$
$122 \mapsto 171$	$124 \mapsto 114$	$125 \mapsto 173$	$126 \mapsto 115$	$127 \mapsto 177$	$129 \mapsto 175$
$130 \mapsto 181$	$132 \mapsto 179$	$133 \mapsto 185$	$135 \mapsto 183$	$136 \mapsto 189$	$138 \mapsto 187$
$139 \mapsto 193$	$141 \mapsto 191$	$142 \mapsto 197$	$144 \mapsto 195$	$145 \mapsto 201$	$147 \mapsto 199$
$148 \mapsto 205$	$150 \mapsto 203$	$151 \mapsto 209$	$153 \mapsto 207$	$154 \mapsto 213$	$156 \mapsto 211$
$157 \mapsto 217$	$159 \mapsto 145$	$160 \mapsto 148$	$161 \mapsto 219$	$162 \mapsto 0$	$163 \mapsto 2$
$164 \mapsto 231$	$166 \mapsto 232$	$168 \mapsto 233$	$169 \mapsto 222$	$170 \mapsto 225$	$171 \mapsto 149$
$172 \mapsto 215$	$173 \mapsto 224$	$175 \mapsto 135$	$178 \mapsto 108$		

References

[1] Vladimir Reshetnikov and ProveIt contributors. *ProveIt*, comparison snapshot `3ecd6a6cf53d47519228328567f45fb7ee9781d9`, 25 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/3ecd6a6cf53d47519228328567f45fb7ee9781d9.

[2] ProveIt contributors. *Con(ZFC) in Diophantine form*, delivered manuscript index and comparison notes, at the same snapshot. https://github.com/VladimirReshetnikov/ProveIt/tree/3ecd6a6cf53d47519228328567f45fb7ee9781d9/Computability/DiophantineConsistency/con-zfc-diophantine-forms.

[3] ProveIt contributors. *Latest proposal review*, at the same snapshot. https://github.com/VladimirReshetnikov/ProveIt/blob/3ecd6a6cf53d47519228328567f45fb7ee9781d9/Computability/DiophantineConsistency/PROPOSAL_REVIEW_LATEST.md.

[4] *Centered Residues*, incoming archive, 25 October 2026. Compact component, including the retained arithmetic proofs and indexed constant recipe. Original notices are preserved in this package. https://github.com/VladimirReshetnikov/ProveIt/blob/3ecd6a6cf53d47519228328567f45fb7ee9781d9/docs/incoming/ConZFC_Centered_Residues_2026-10-25.zip.

[5] *Residual Bases and Denominator Engineering*, incoming archive, 25 October 2026. In particular `compact/PROOF.md`, Section 8, records the rejected $Y = qs$ first-order approach. https://github.com/VladimirReshetnikov/ProveIt/blob/3ecd6a6cf53d47519228328567f45fb7ee9781d9/docs/incoming/ConZFC_Residual_Bases_Denominator_Engineering_2026-10-25.zip.

[6] *Coprime Products and Positive Centers*, incoming archive, 25 October 2026. The partial-scale predecessor motivates delayed recovery of divisibility in a rational carrier. https://github.com/VladimirReshetnikov/ProveIt/blob/3ecd6a6cf53d47519228328567f45fb7ee9781d9/docs/incoming/ConZFC_Coprime_Products_and_Positive_Centers_2026-10-25.zip.

[7] *Common Remainders and Radix 47*, incoming archive, 25 October 2026. The direct component is retained unchanged in `direct_quartics/upstream_common/`, with all 88 copied files pinned. https://github.com/VladimirReshetnikov/ProveIt/blob/3ecd6a6cf53d47519228328567f45fb7ee9781d9/docs/incoming/ConZFC_Common_Remainders_Radix47_2026-10-25.zip.

[8] *Graph Digits*, delivered package 139, compact component. The 87/323 and 88/318 points used in the comparison. https://github.com/VladimirReshetnikov/ProveIt/tree/3ecd6a6cf53d47519228328567f45fb7ee9781d9/Computability/DiophantineConsistency/con-zfc-diophantine-forms/139-graph-digits/compact.

[9] ProveIt contributors. The retained interfaces `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. See `direct_quartics/upstream_common/upstream/round17/architecture/proofs/inherited/calibration/` and its exact upstream source pins. The corresponding native-route interfaces and provenance are in `compact/parent/proof_sources/`.